Week 3 — Distributions and associations

Business Forecasting

Krzysztof Zaremba

My code (.R) Notes

Where we are

Describing data → Reasoning under uncertainty → Regression → Time series forecasting

Describing data, part 2 — and the last of it. Last week boiled a variable down to a few numbers: where it sits and how spread out it is. This week we stop summarising and look at the whole distribution — then put two variables side by side and measure how they move together. Next week we stop describing and start inferring.

Distributions

How is the data *spread out*? Now picture the *whole* distribution.

How can we describe and visualize a distribution?

Last week we squeezed a variable down to a handful of numbers: where it sits and how spread out it is. This week we want the whole picture — how the values are spread along the line, and how to draw that.

Week 2 already gave us one recipe: the frequency table. Count how often each value occurs, then plot the shares. `age` is a number, not a label — but nothing stops us doing it anyway.

Does the same recipe work for every numeric variable?

Exercise 1 — Try the recipe on both

Build the relative-frequency plot of `Health_data$age` the Week-2 way. Then run exactly the same code on `Health_data$weight`. Compare the two pictures and say what broke.

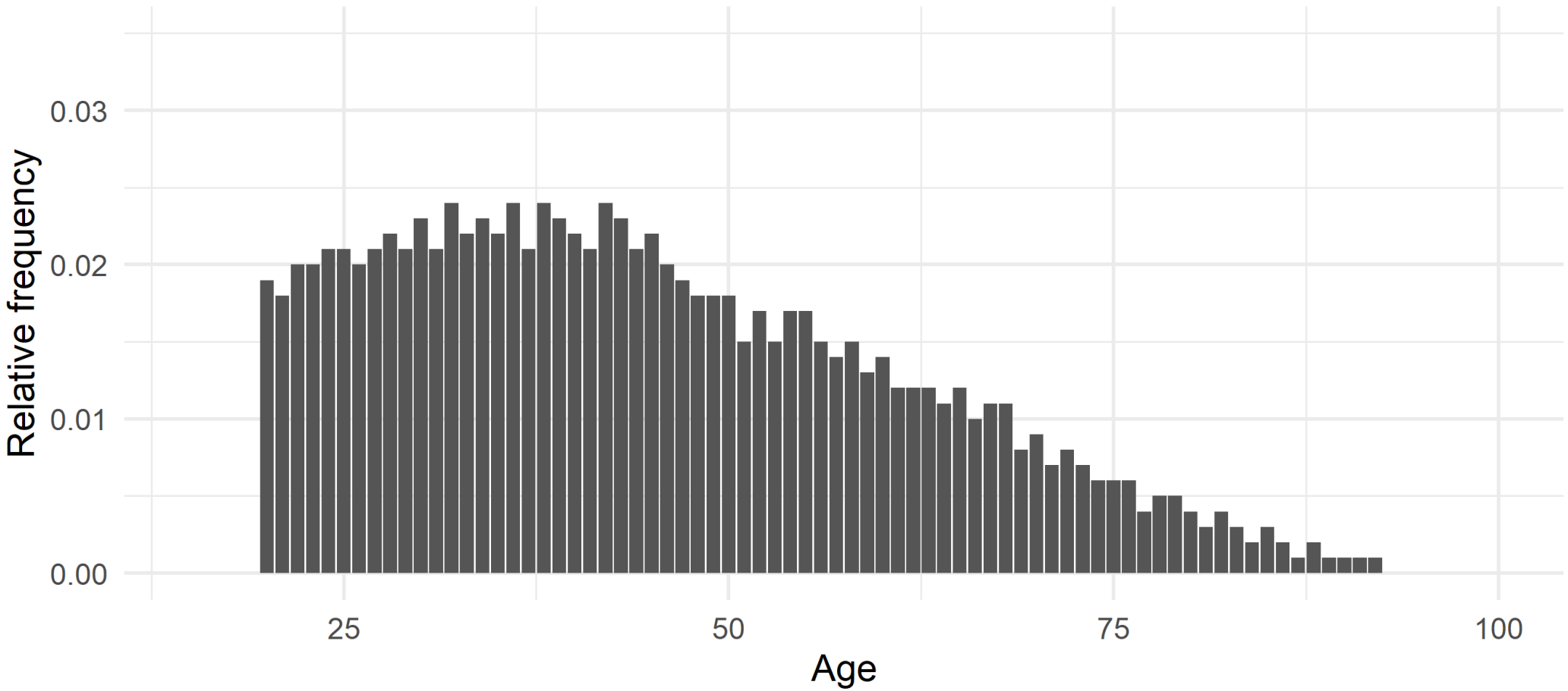
Exercise | Start Over | Show Hint | Run Code

```
# age first
p_age <- prop.table(table(Health_data$age))

# now the identical recipe on weight
```

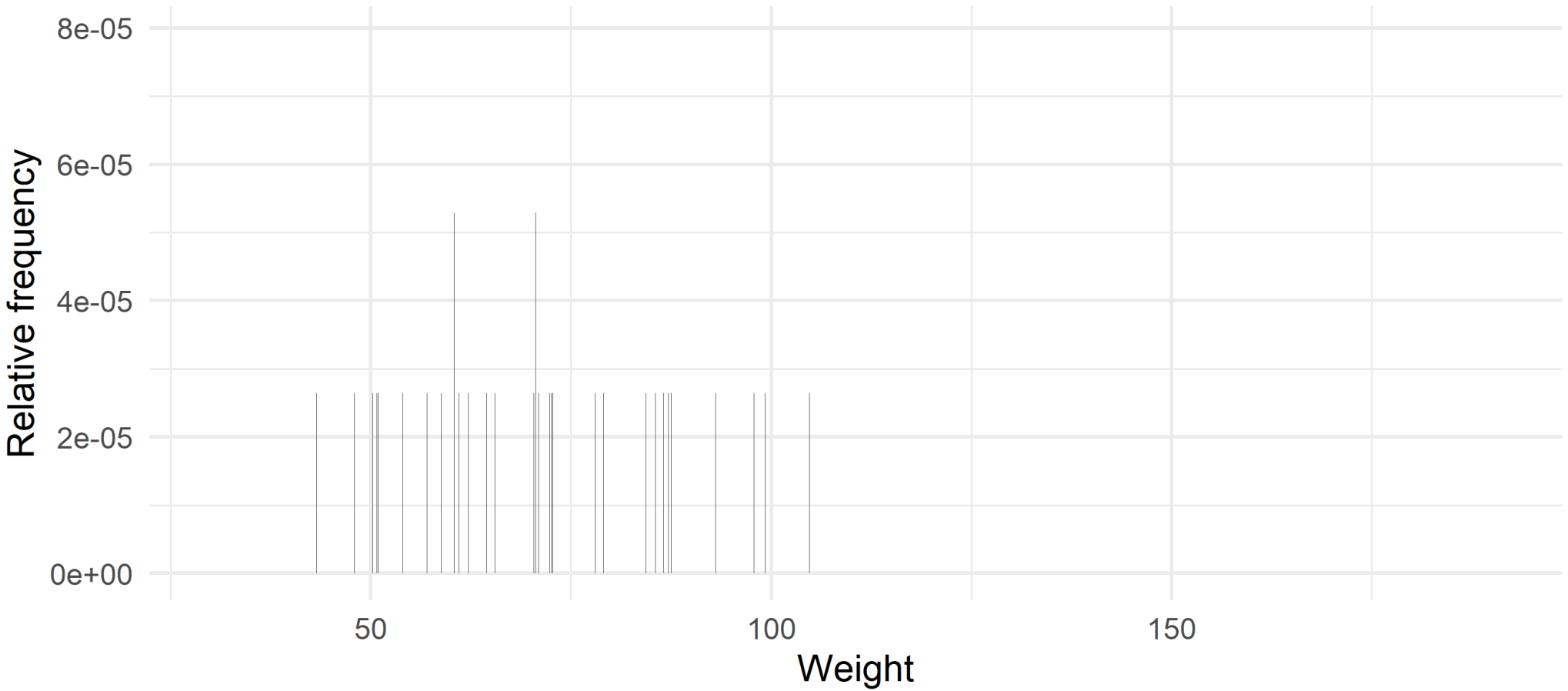
Frequency distribution of a numeric variable

Compare age in the adult population (20+) — how often does each value occur?



Continuous variables: the problem

- The same recipe on a continuous variable like weight:



- Most values never repeat, so each has a tiny relative frequency — the plot is useless.

Histograms

Solution: Group similar values together

- Construct intervals and show how many observations are in a given interval

Process

1. Decide how many intervals
2. And how wide they are
3. Then calculate the absolute and relative frequencies of each interval
4. Plot it with bars

Histograms: choosing bin width

- I want k equal intervals — say $k = 5$ — covering all of `Health_data$weight`, from the smallest value to the largest.
- Two things are needed to lay them out: how much ground the data covers, and how much of that ground one interval gets.

Exercise 2 — How wide is one interval?

Cut `Health_data$weight` into 5 equal intervals. How wide is one of them? One call per line: the two ends first, then the range, then the width.

Exercise | Start Over | Show Hint | Run Code

```
# 1. the largest value of Health_data$weight

# 2. the smallest value of Health_data$weight

# 3. the range: largest minus smallest

# 4. the width of one interval when there are 5 of them
```

Bin width: the numbers

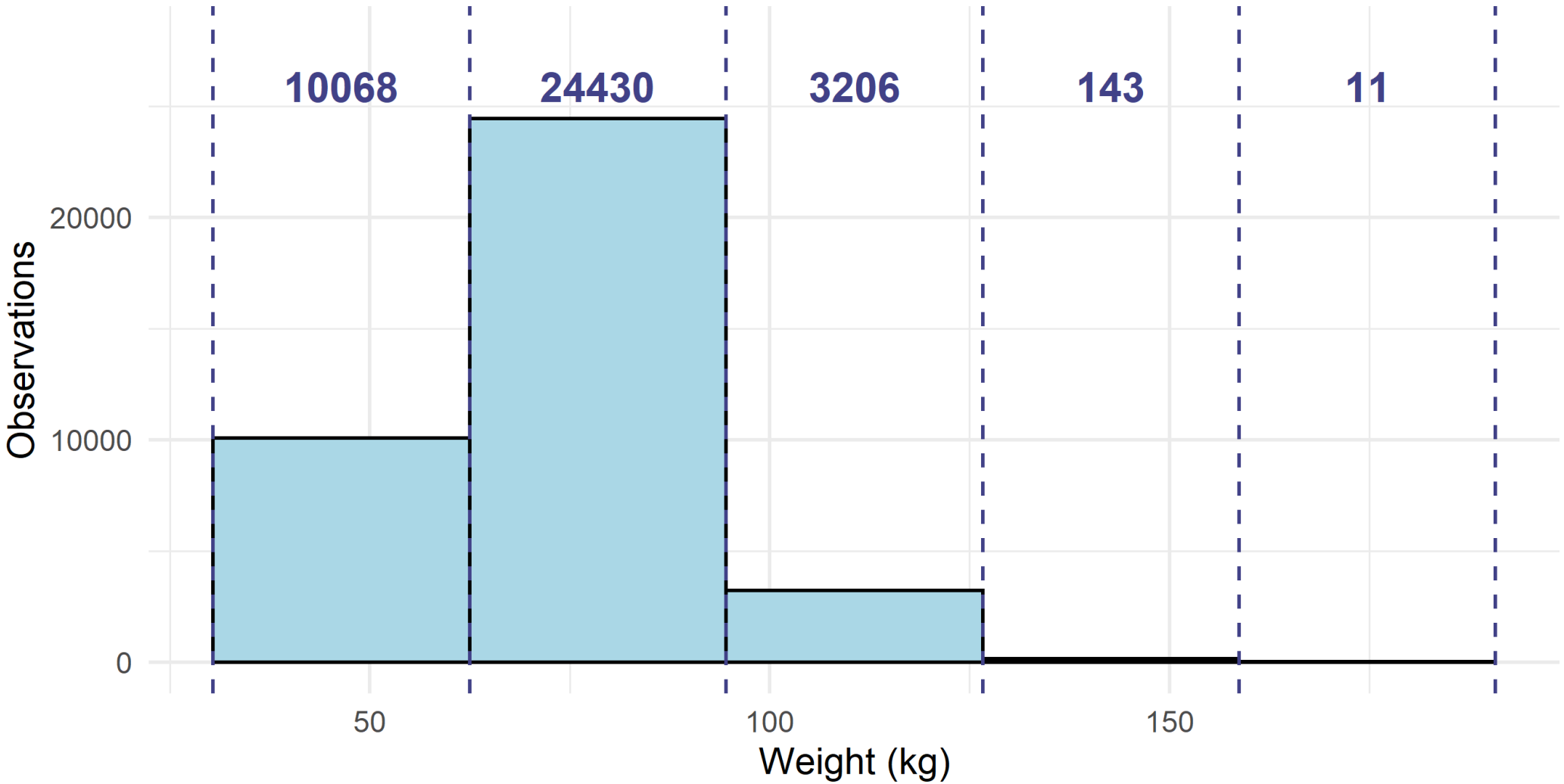
- Range = 190.81 − 30.37 ≈ 160.4 kg.
- Split into 5 equal intervals, each one is ≈ 32 kg wide.
- The first interval starts at the minimum (30.37) and the last ends at the maximum (190.81).

The five intervals, counted

Weight interval (kg)	Observations	Share
30.4 – 62.5	10068	0.266
62.5 – 94.5	24430	0.645
94.5 – 126.6	3206	0.085
126.6 – 158.7	143	0.004
158.7 – 190.8	11	0.000

Two intervals hold 91% of the people; the top two hold 154 between them. Equal widths in kilograms do not mean equal numbers of people.

The same five intervals, drawn



Five bars, one per row of the table. The dashed lines are the interval edges you computed — the bars are just those counts drawn to scale.

Exercise 3 — Draw it in R

Build a histogram of `weight`. Then answer from your own picture, before we show any: roughly where does it peak, and is either tail longer than the other?

Exercise | Start Over | Show Solution | Run Code

```
# Your code here

# peak at about ____ kg ; the longer tail is on the ____
```

Exercise 4 — Break the bins

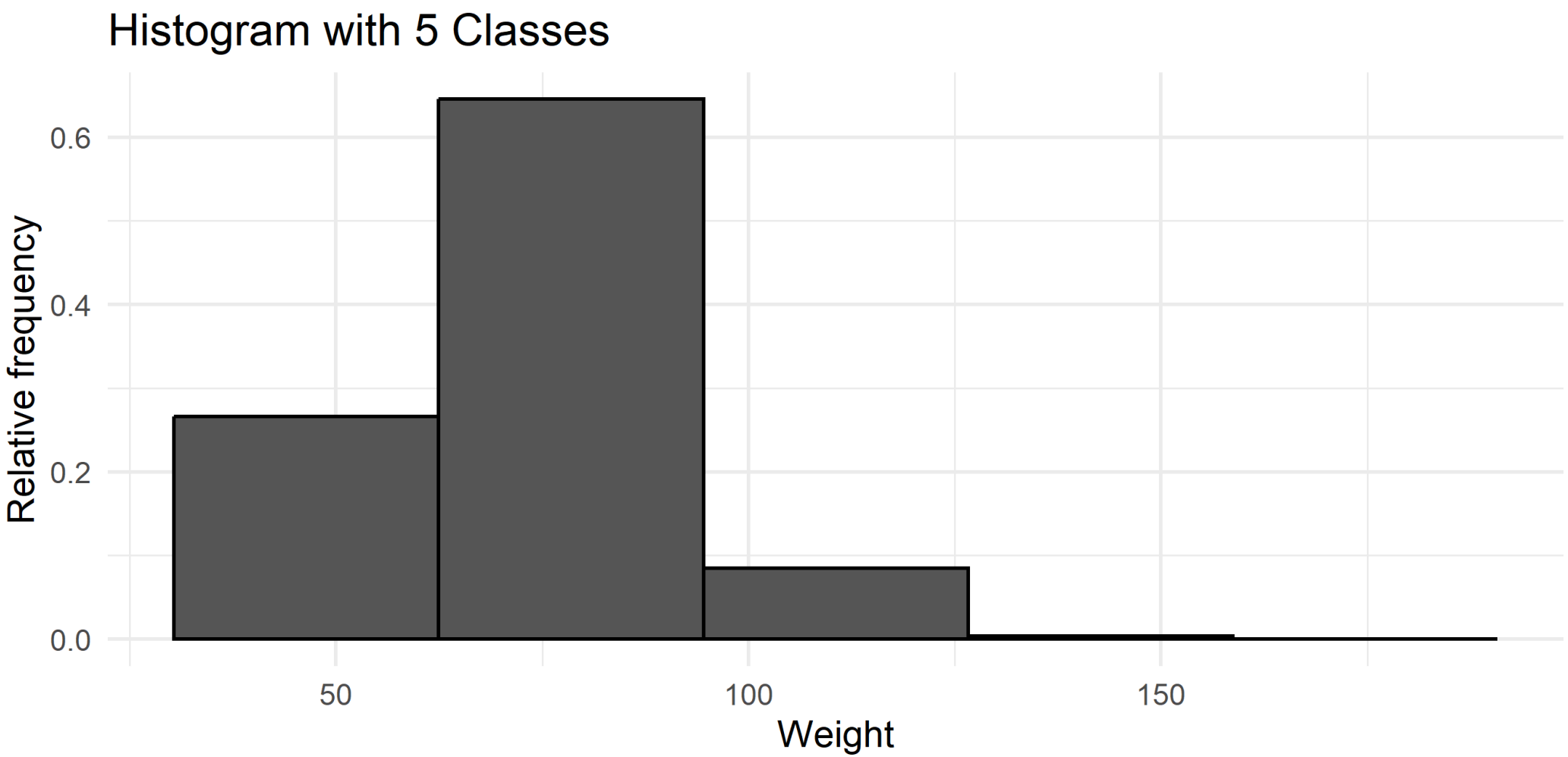
Draw the histogram of `Health_data$weight` four times, with `breaks = 5`, `30`, `100`, `1000`. The number of bins is your choice, not the data's. Look at all four, then answer: which one hides the skew, which one is unreadable noise, and which one would you put in the report?

Exercise | Start Over | Show Hint | Run Code

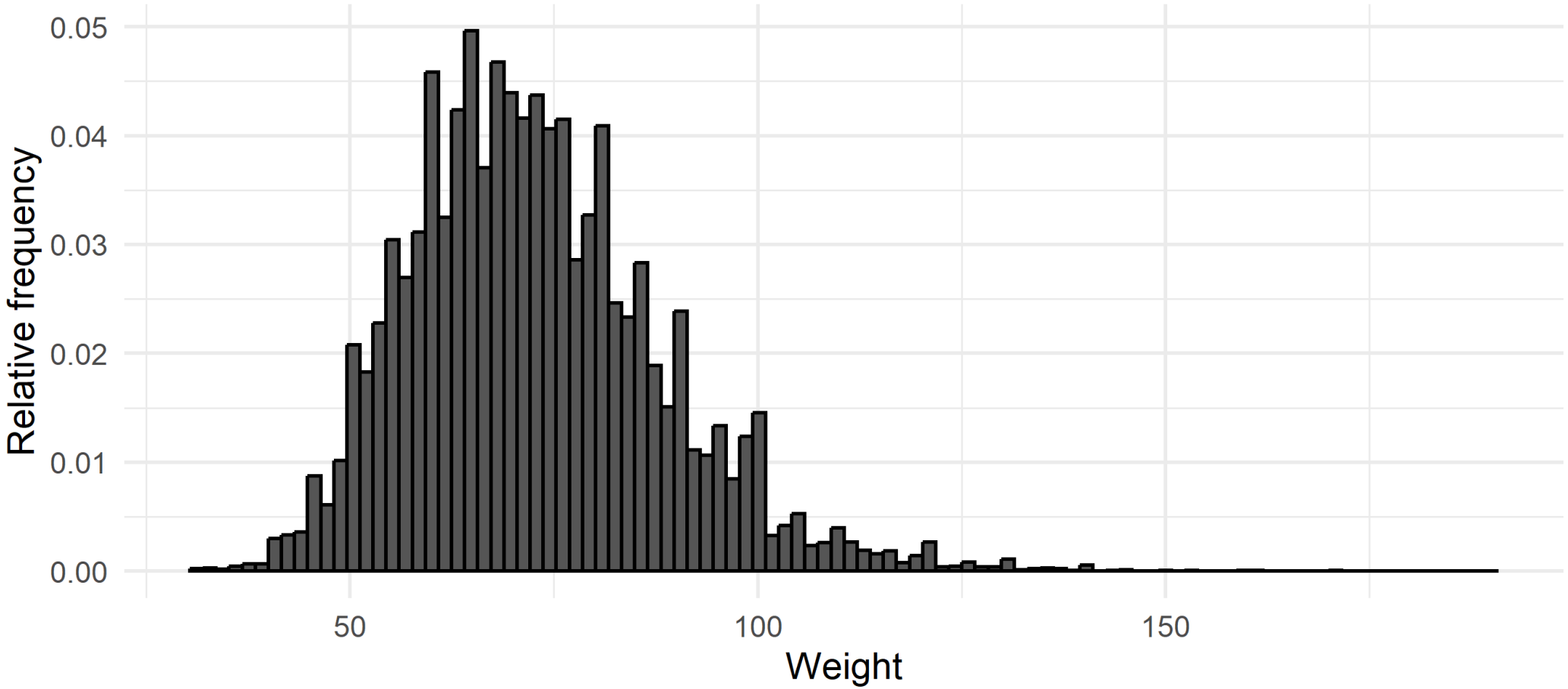
```
1 par(mfrow = c(2, 2))   # four plots in one window
```

Histogram with 5 classes

- Midpoint = center of the interval (center of the bar)



Histogram with 100 classes



- Helps to see the distribution and outliers — but is more always better?
- The bars change every time you change `breaks`. Is there anything underneath that doesn't?

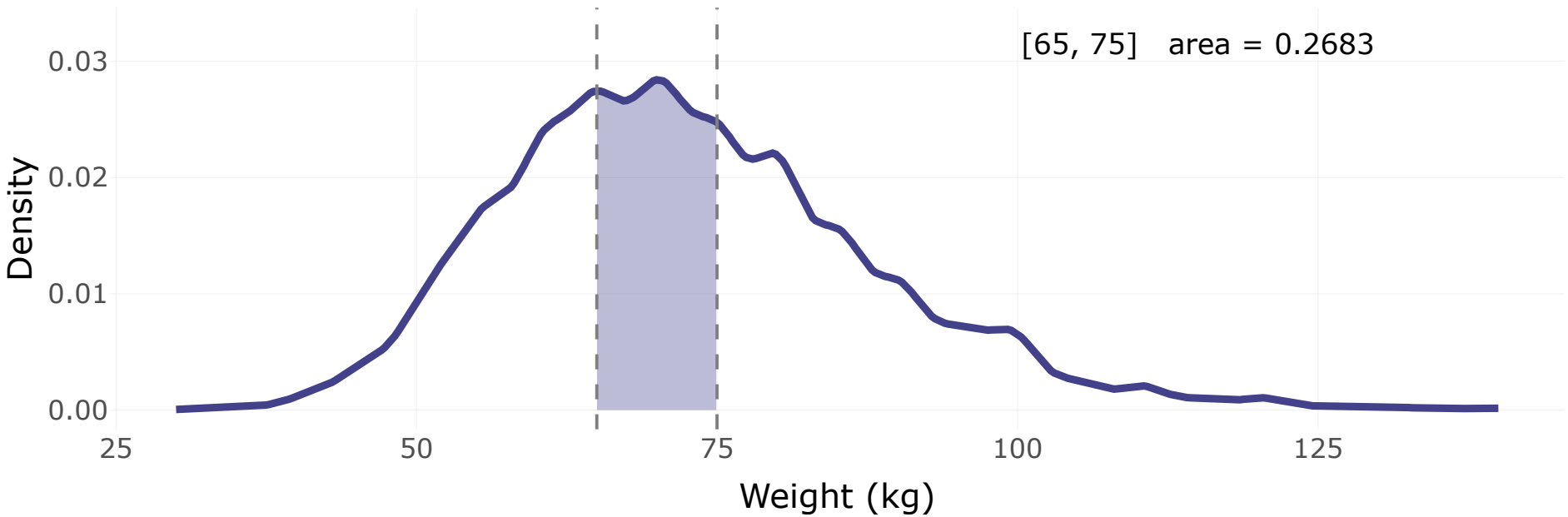
What the bars are approximating

- With smaller and smaller intervals, the histogram tends to a smooth curve.
- The histogram is a *sample* picture; the curve is the population object.
- Mathematically there are two ways to write that curve down: the PDF and the CDF.

The PDF — where the values pile up

$$P(a \le X \le b) = \int_a^b f(x)\,dx$$

Intuition: the height of the curve says where values *pile up*. Probability is the area under it, so a single point has probability zero — the area of a line is zero. The whole curve integrates to 1.



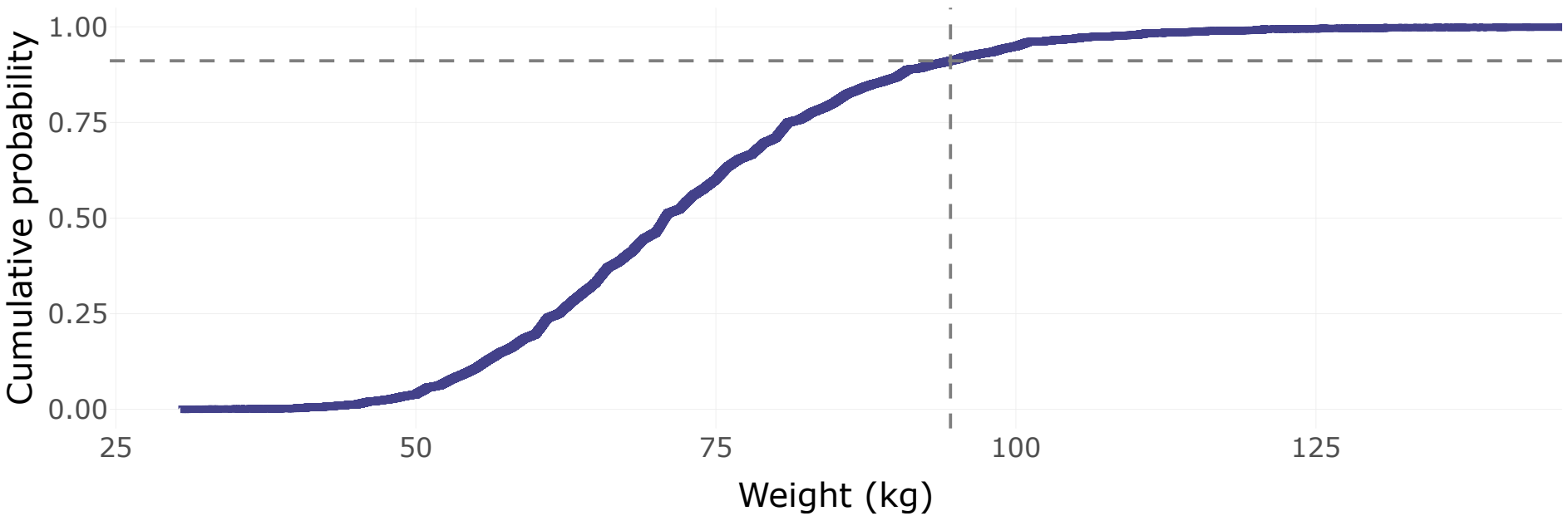
Example: the shaded area is $P(65 \le X \le 75)$ — the share of adults weighing between 65 and 75 kg, which in this sample is 27%. Hover to read the density at any weight.

Why does a single weight have probability zero? See it shaded.

The CDF — how much is at or below

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)\,dt$$

Intuition: walk up the axis and keep a running total of everyone you have passed. It starts at 0, never goes down, and ends at 1. The PDF is the *slope* of this curve; the CDF is the *area* under the PDF so far.



Example: $F(94.55) = 0.911$ — 91% of adults weigh 94.55 kg or less. Read any percentile straight off this curve: pick a share on the vertical axis, cross to the curve, drop to the weight.

Where does that curve come from when you only have a sample? Sort everyone and count.

Describing a distribution in words

You now have three ways to *show* a distribution — a histogram, a PDF, a CDF. But in a report you have to say what it looks like, in words a manager understands.

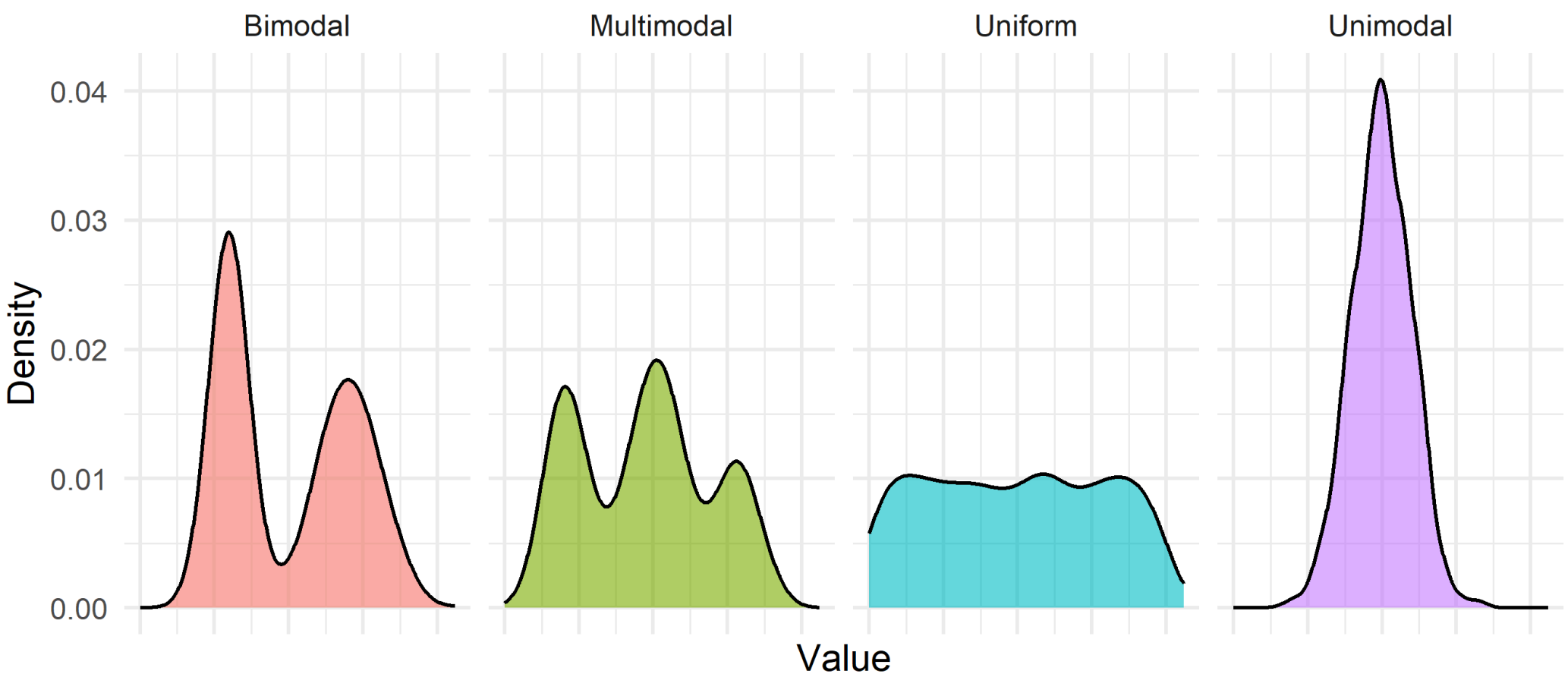
Two words do most of the work:

- Modality — how many peaks does it have?
- Skewness — is one tail longer than the other?

They are worth learning precisely, because they decide something practical: which centre and which spread measure you are allowed to report.

Distribution Shapes: Modality

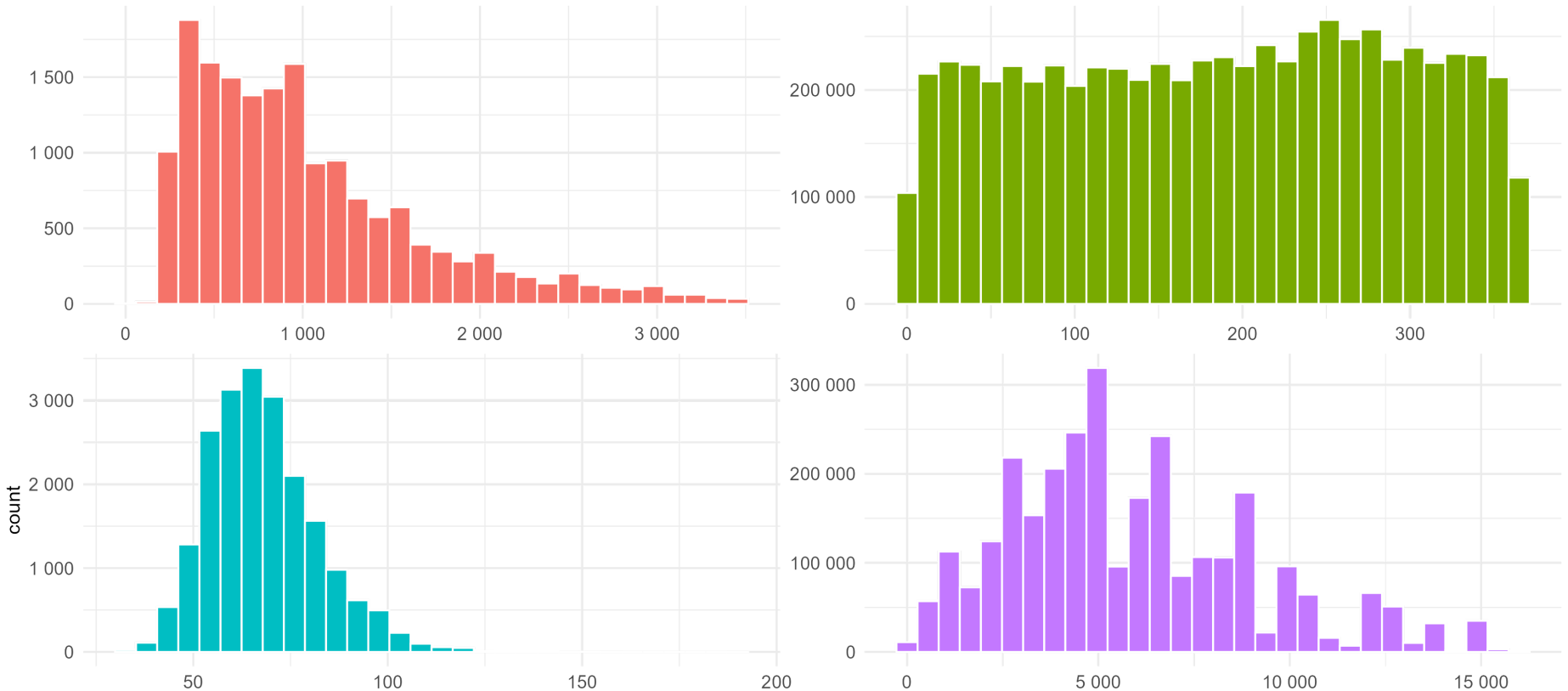
How many “peaks” does the distribution have?



Which is uniformly distributed?

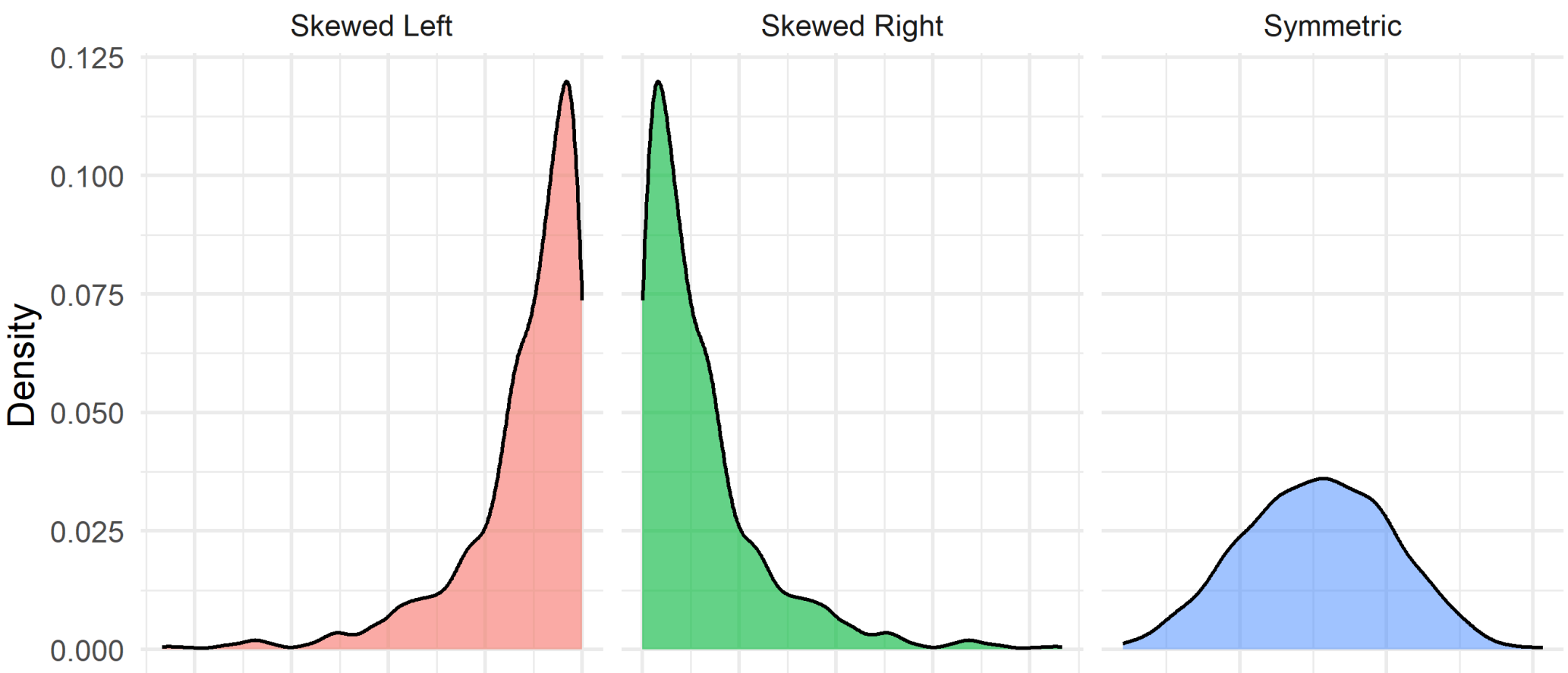
1. Weights of Adult Females
2. Salaries in Mexico
3. Airbnb prices in CDMX
4. Birthdays of Classmates (day of the month)

Which distribution is which?



Distribution Shapes: Skewness

Is the tail longer on the left or the right?



Reading skew without looking

Which is bigger for a right-skewed variable — the mean or the median?

Exercise 5 — Predict, then check

You have already drawn the histogram of `Health_data$weight`. Commit to a guess first: is its mean above or below its median? Write it down, then compute both.

Then repeat for `Health_data$age`, and draw the histogram to see whether the two numbers told you the truth about the shape.

Exercise | Start Over | Show Solution | Run Code

```
1 # my guess for weight: mean is ____ the median
2 # my guess for age:    mean is ____ the median
```

If the mean is clearly larger than the median, the distribution is…

Right-skewed — the long right tail pulls the mean up above the median.

Shape in R: the analyst's move

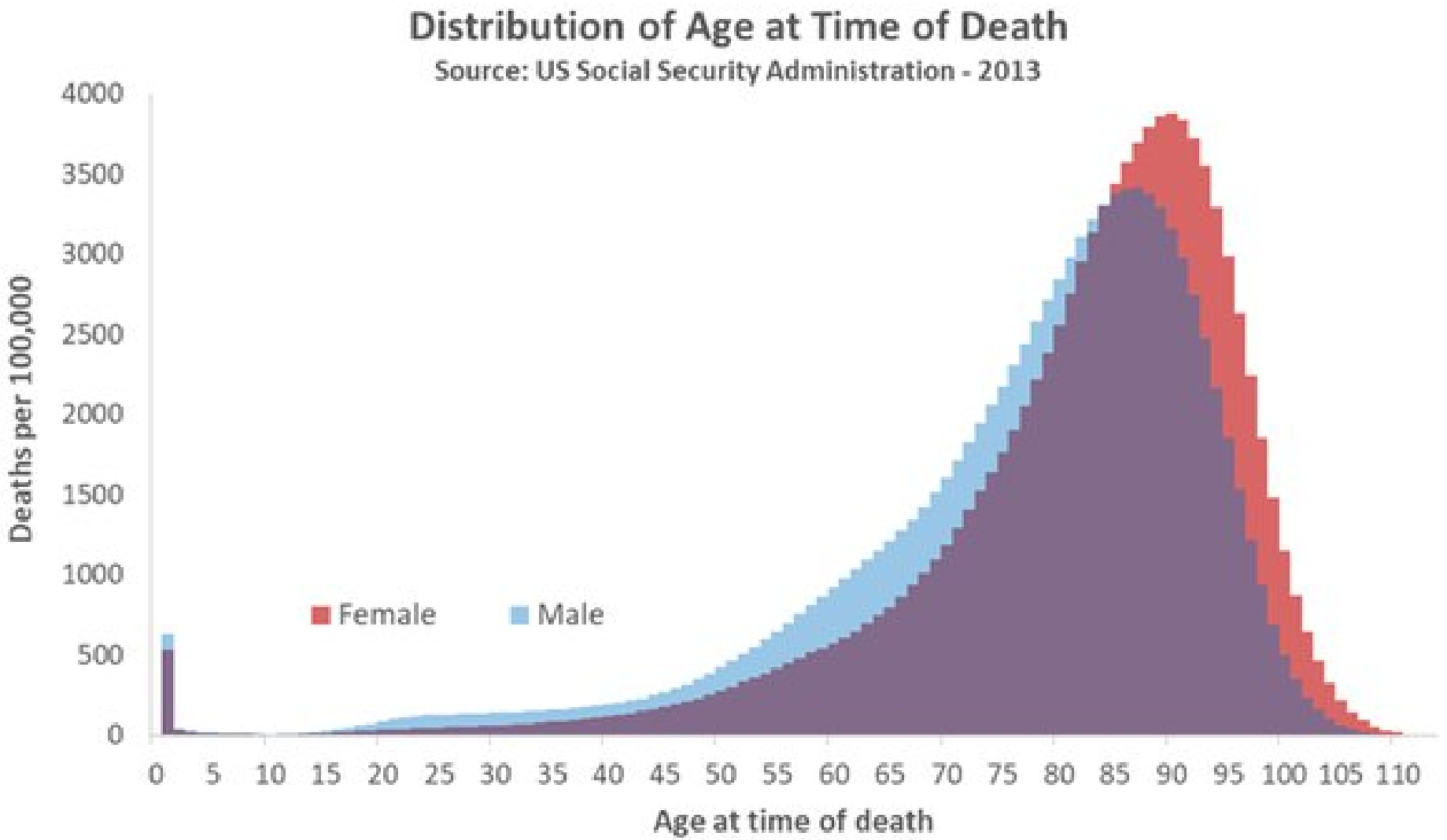
- The whole job of reading a distribution: plot it, then name its shape.
- Modality — count the peaks: one (unimodal), two (bimodal), many.
- Skewness — which tail is longer: left, right, or symmetric.
- In R: `hist(df$var)`, then look at it.

That is enough to answer questions like:

- *Is the "average" salary we quote in the job ad typical, or is it pulled up by a few executives?* — right-skewed, so quote the median.
- *Do our customers split into two groups?* — two peaks means two segments, and one average describes neither.
- *How much stock covers a busy day?* — a long right tail means the average day is a bad planning number.

Age at death (a left-skewed variable)

You cannot price a life policy without the whole distribution of age at death — the premium depends on how often people die young, not on the average.



Comparisons and associations

One variable at a time is only half the job. Business questions are almost always about two variables together.

From one variable to two

So far every tool described one variable: its centre, its spread, its whole distribution.

But the questions that get asked at work are comparisons:

- Do customers in rural areas churn more than customers in cities?
- Do people who saw the ad spend more than people who did not?
- Does an extra year of schooling come with higher pay?

Each of these puts two variables side by side. The tool you need depends on what *kind* of variables they are.

Which tool for which pair

First variable	Second variable	Tool
Categorical	Categorical	Contingency table, stacked barplot
Quantitative	Categorical	Group means, boxplots, compared distributions
Quantitative	Quantitative	Scatterplot, covariance, correlation

We take them in that order.

Throughout: descriptive comparisons only. Deciding whether a gap is large enough to be real is inference — that starts in Week 4.

Two categorical variables

You sell air conditioners. Where do you expand next — big cities or small towns?

`households` — 2,000 Mexican households, surveyed once in 2017:

- `locality_size` — `"Large_city"`, `"Medium_city"`, `"Small_town"`, `"Very_small_village"`
- `uses_electricity_cooling` — `TRUE` if the household cools with electricity

The households that already have cooling are not your market, so you need one number per locality: what share is already covered?

Which locality is most likely to have cooling? Which is least likely?

Exercise 9 — Count them yourself

Build the table that cross-classifies every household in `households` by `locality_size` and `uses_electricity_cooling`. Then read off: how many large-city households cool electrically, and how many village households do?

Exercise | Start Over | Show Hint | Run Code

```
1 # cross-classify locality_size and uses_electricity_cooling
```

Joint frequencies

	No cooling	Electric cooling
Large_city	369	514
Medium_city	138	173
Small_town	179	165
Very_small_village	272	190

- The inner cells are joint frequencies: how many households are in *both* categories at once.
- Divide by the sample size and they become joint relative frequencies — the share of the whole sample in each cell.

	No cooling	Electric cooling
Large_city	0.184	0.257
Medium_city	0.069	0.086
Small_town	0.090	0.082
Very_small_village	0.136	0.095

- 25.7% of all households are large-city *and* cooled; 9.5% are village *and* cooled.
- These still cannot be compared to each other, for the same reason as the counts.

Joint, marginal, conditional

$$P(C \mid L = \text{Village}) = \frac{P(C \cap L = \text{Village})}{P(L = \text{Village})}$$

In words: rescale each row by its own group size, so every row answers “out of 100 households like this, how many cool electrically?”

Exercise 10 — Rescale each row

Turn your table of counts into shares within each locality, so the four rows can be compared. Then say where cooling is most common and where it is least.

Exercise | Start Over | Show Hint | Run Code

```
1 # start from the counts, then rescale row by row
2 tab <- table(households$locality_size, households$uses_electricity_cooling)
```

The contingency table

	No cooling	Electric cooling
Large_city	0.418	0.582
Medium_city	0.444	0.556
Small_town	0.520	0.480
Very_small_village	0.589	0.411

- Each row now sums to 1, and the four rows are directly comparable.
- A table of conditional shares like this is called a contingency table.
- Whether a gap this size is more than sampling noise is a question for Week 4 — here we only describe it.

Read the rows, never the columns

Row sums are meaningful

$$P(C \mid \text{Village}) + P(\text{no } C \mid \text{Village}) = 1$$

Every village household either cools electrically or does not.

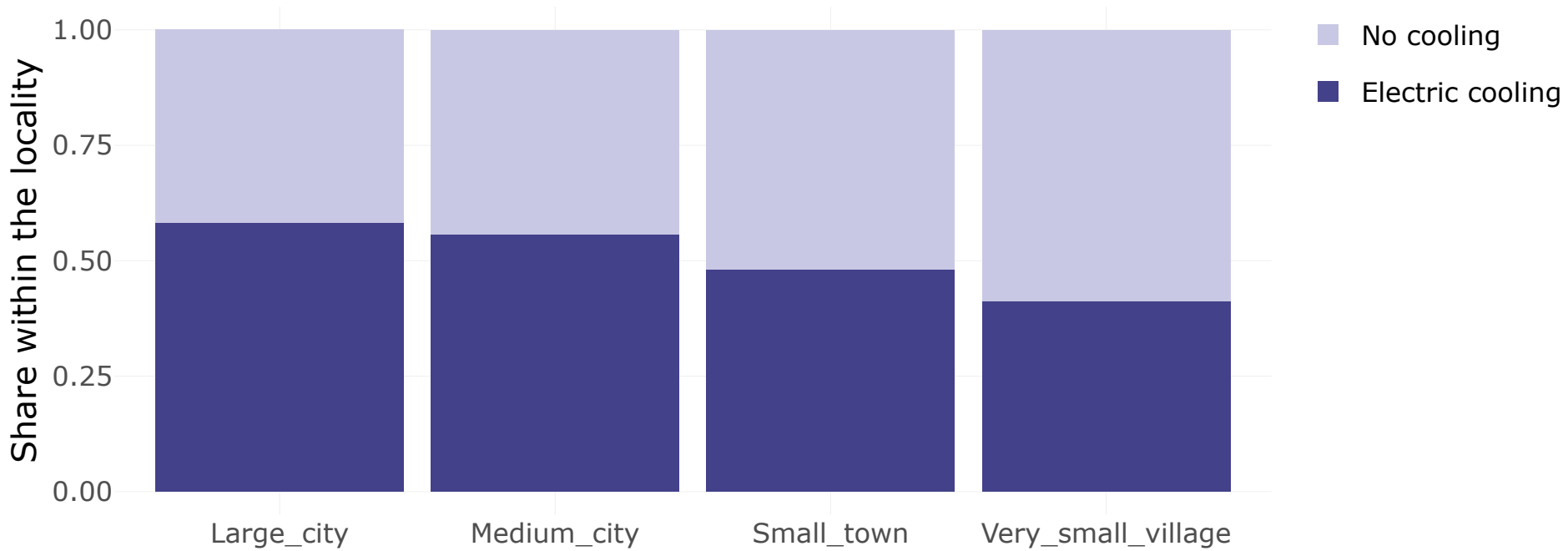
Column sums are meaningless

$$P(C \mid \text{Village}) + P(C \mid \text{Large city}) = ?$$

These are shares of two *different* groups. Adding them answers nothing.

Which way you condition is a choice, and it changes the question. `margin = 1` asks "of village households, how many cool?"; `margin = 2` asks "of cooled households, how many are in villages?" Both are legitimate; only one answers the question you asked.

Seeing it: the stacked barplot



- Each bar is one locality, rescaled to full height — so the bars compare *rates*, not how many households the survey happened to collect in each place.
- The pale segment is the part of each market still open: largest in the villages, at 59%. Whether that makes villages the *best* place to expand is a different question — it also depends on what those households can pay.

Exercise 11 — Where is the recycling market?

A packaging firm wants to know where households already sort their waste. `households$separates_trash` is `TRUE` if the household separates its trash. Repeat the whole procedure: cross-classify it with `locality_size` and rescale within each row.

Households in localities of which size have the highest probability of recycling? Report the four rates, and say whether this association runs the same way as cooling did.

Exercise | Start Over | Show Hint | Run Code

```
1 # same two steps as before, on separates_trash
```

What that does *not* establish

Villages recycle more. It is tempting to conclude that living in a village makes people sort their waste.

But village households differ in many other ways at once: house size, yard space, whether there is a collection service at all, income, age. The table cannot separate any of those from the locality itself.

We come back to this at the end of the section — correlation is not causation.

One quantitative, one categorical

One variable is a number, the other is a label.

`salaries` — 20,000 workers from the Mexican labour force survey (ENOE, 2022):

- `income` — monthly income in pesos
- `female` — `1` for women, `0` for men

Is the mean income of women lower than the mean income of men? By how many pesos? And is the spread of income the same in both groups?

Three ways to compare two groups

- Compare the means — one number for each group.
- Compare the histograms — the full distribution of each group.
- Compare the boxplots — both groups in one picture.

We do all three, in that order. Each one shows something the one before it hides.

Exercise 12 — Compare the means

Compute the mean of `salaries$income` for the workers with `female == 1`, and again for the workers with `female == 0`.

Which group has the higher mean? How large is the gap in pesos?

Exercise | Start Over | Show Hint | Run Code

```
1 # one group at a time
```

Method 1: compare the means

Group	n	Mean	Median	SD
Women	8536	7401	6235	5787
Men	11464	9746	8000	7178

- The mean gap is 2,344 pesos. The median gap is even larger in relative terms: 6,235 against 8,000.
- But a mean is one number per group. It cannot tell you whether the two groups overlap, or whether the gap comes from a few very high earners.

Exercise 12b — Compare the histograms

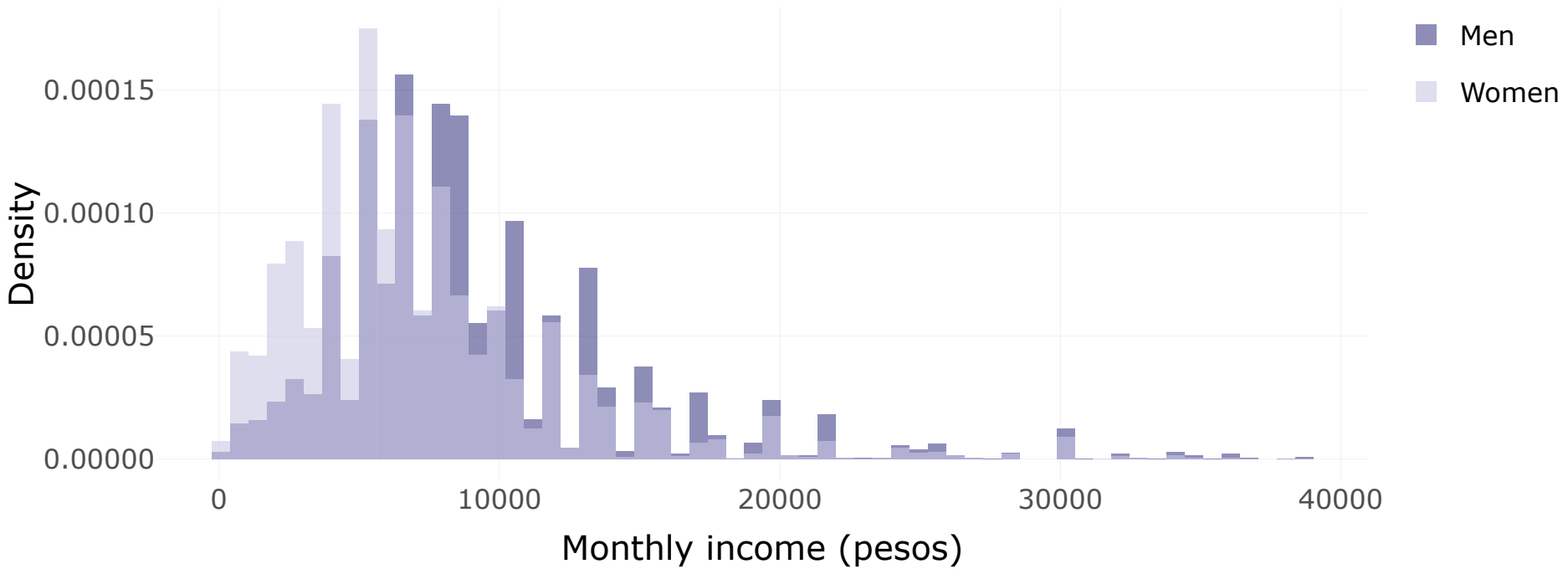
Draw one histogram of `salaries$income` for the women and another for the men.

Do the two groups overlap? Where does each one have most of its mass?

Exercise | Start Over | Show Hint | Run Code

```
1 # one histogram per group
```

Method 2: compare the histograms



- Both groups are drawn as densities, because there are more men than women in the sample. Raw counts would confuse a size difference with an income difference.
- Now you can see what the means hid: the two distributions overlap heavily. Many women earn more than many men. A gap between averages is a statement about groups, never about any one person.

Exercise 12c — Compare the boxplots

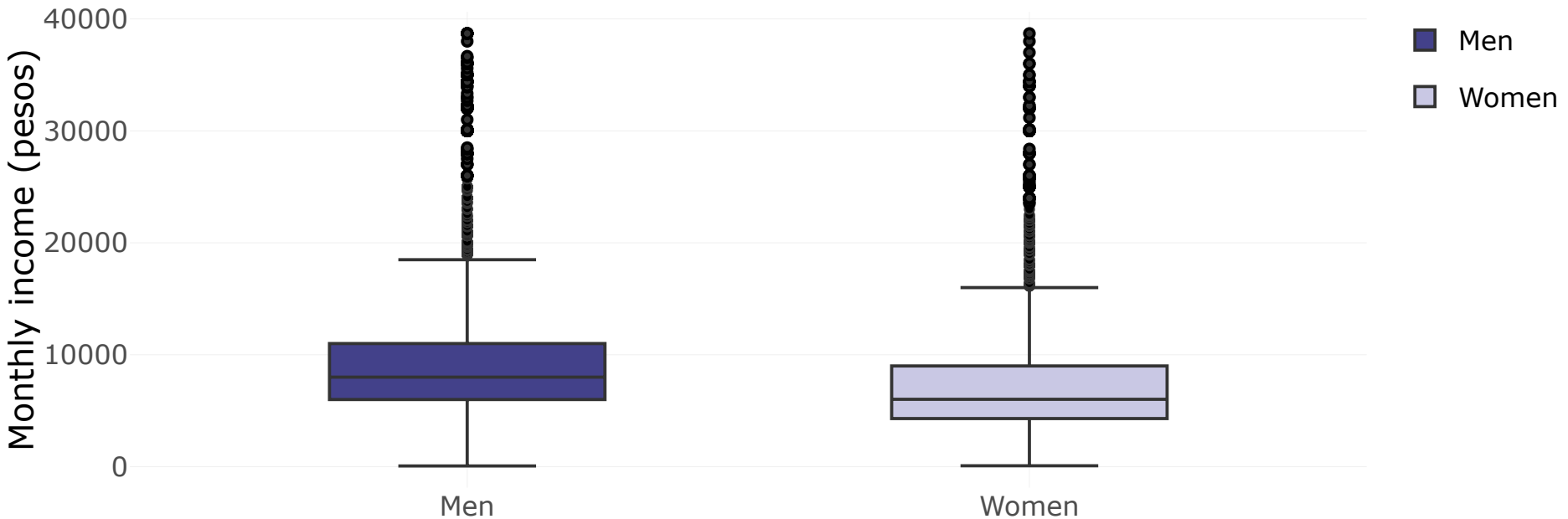
Put both groups in one picture: a boxplot of `salaries$income` split by `salaries$female`.

Which group has the higher median? Which has the wider box?

Exercise | Start Over | Show Hint | Run Code

```
1 # one plot, both groups
```

Method 3: compare the boxplots



- One picture, both groups. It carries the median, both quartiles and the outliers at once, which is why it is the standard way to report this comparison.
- What it loses is shape: a boxplot cannot show you that both distributions have a long right tail. That is what the histograms were for.

Two quantitative variables

Both variables are numbers, so there are no groups to split into.

`spending` — 10,000 Mexican households from ENIGH 2018 (INEGI's income and expenditure survey). Everything is per person, so households of different sizes can be compared:

- `food_pc` — pesos spent on food last month, per person
- `income_pc` — household income that month, per person
- `members` — how many people live in the household

You are planning a supermarket's product mix. Which tells you more about what one person spends on food — the income per person, or the number of people in the household?

The two are measured in different units: pesos and people.

Exercise 13 — Put them on the axes

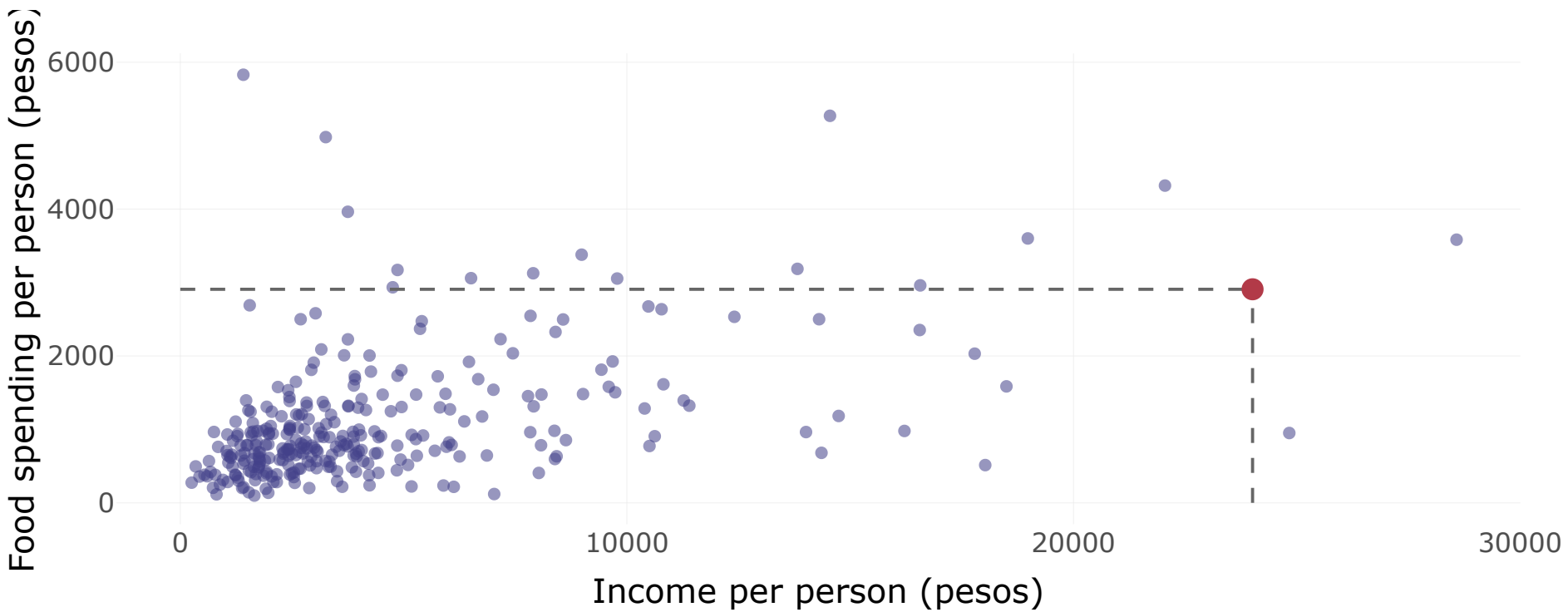
Draw the scatterplot of `spending$food_pc` against `spending$income_pc`.

Does the cloud tilt up or down? Can you read the strength of the relationship off the picture?

Exercise | Start Over | Show Hint | Run Code

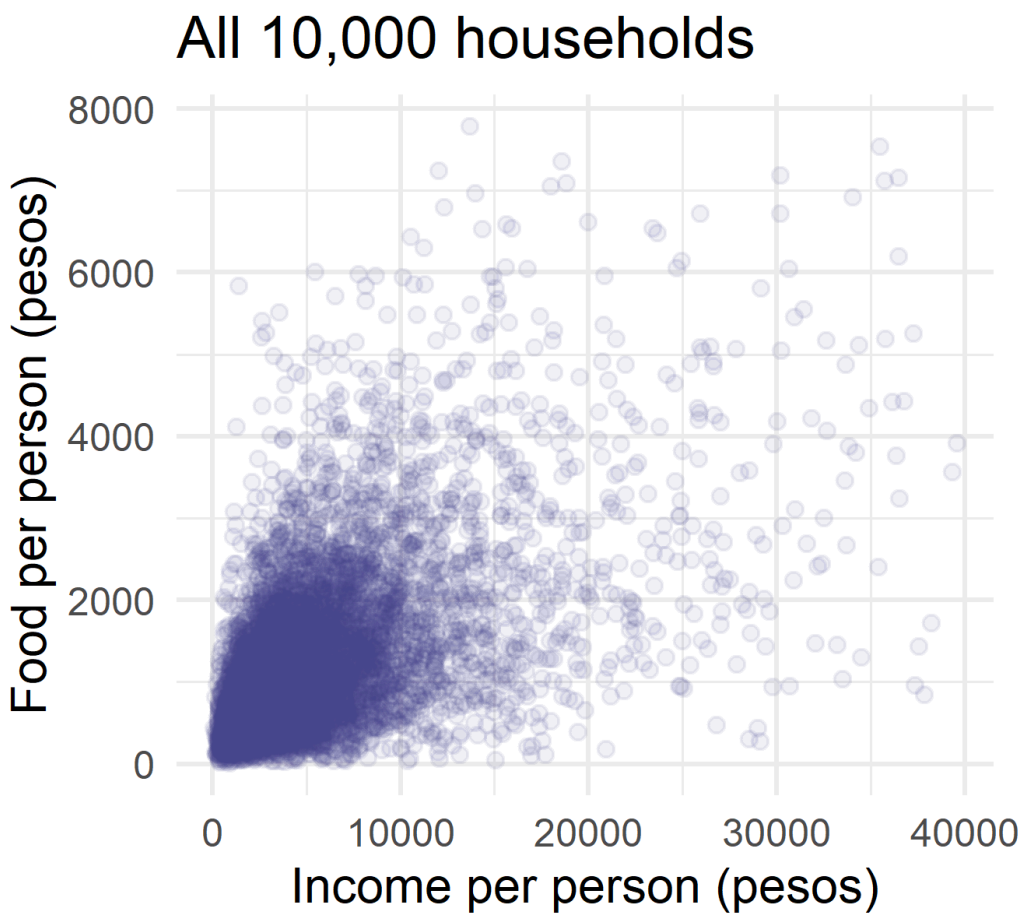
```
1 # income per person on the horizontal axis, food per person on the vertical
```

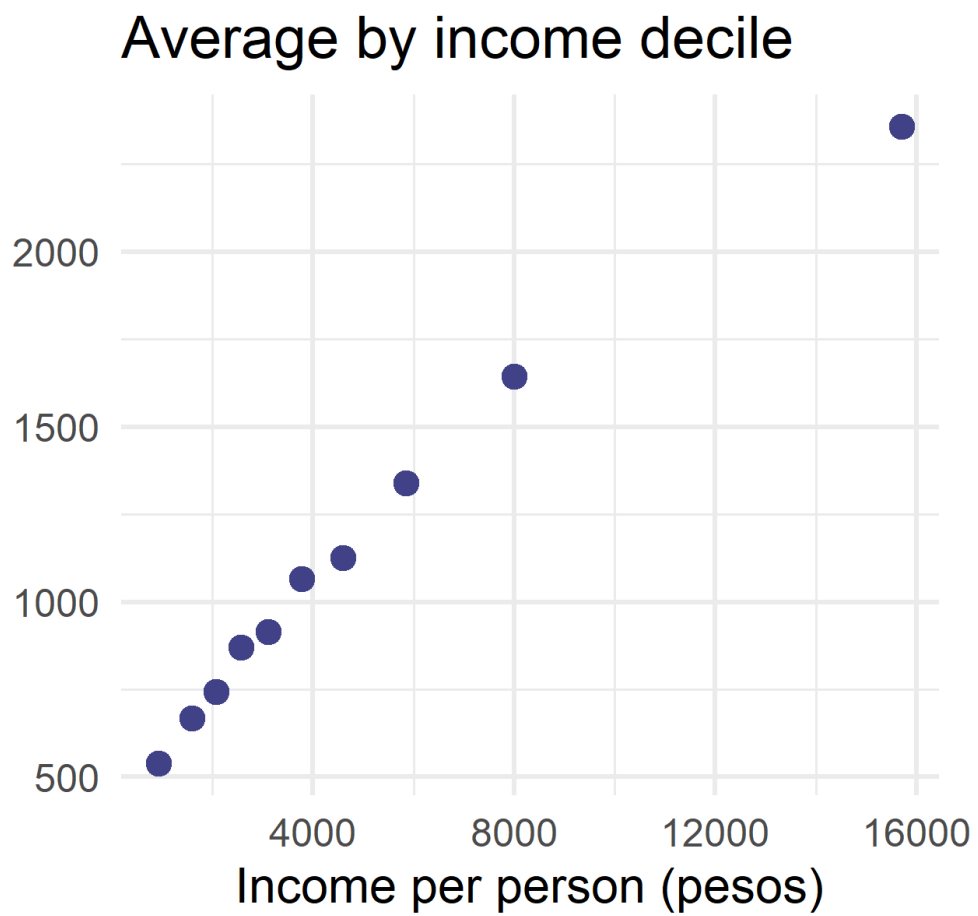
What a scatterplot is



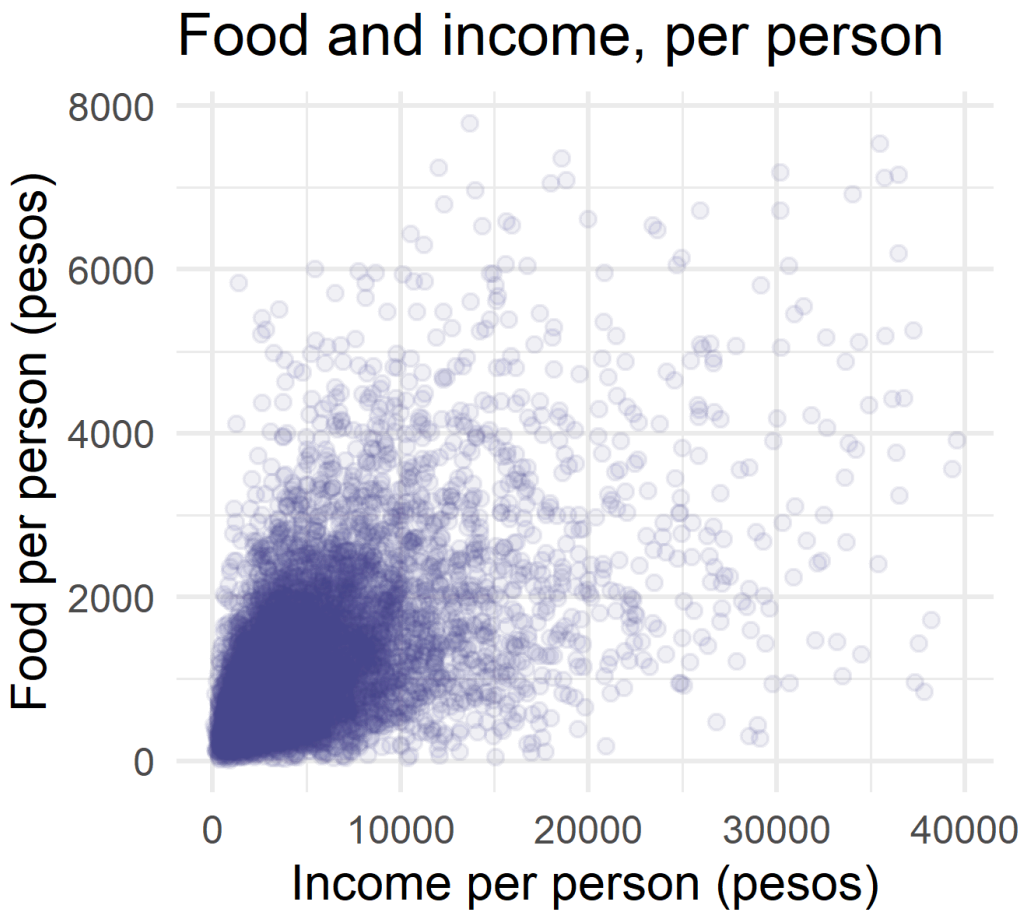
- 300 of the households, so the picture can be read. One dot is one household, placed at its own pair (x_i, y_i) — income per person on the horizontal dashed line, food per person on the vertical one.
- The whole sample is the set of pairs $\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$.

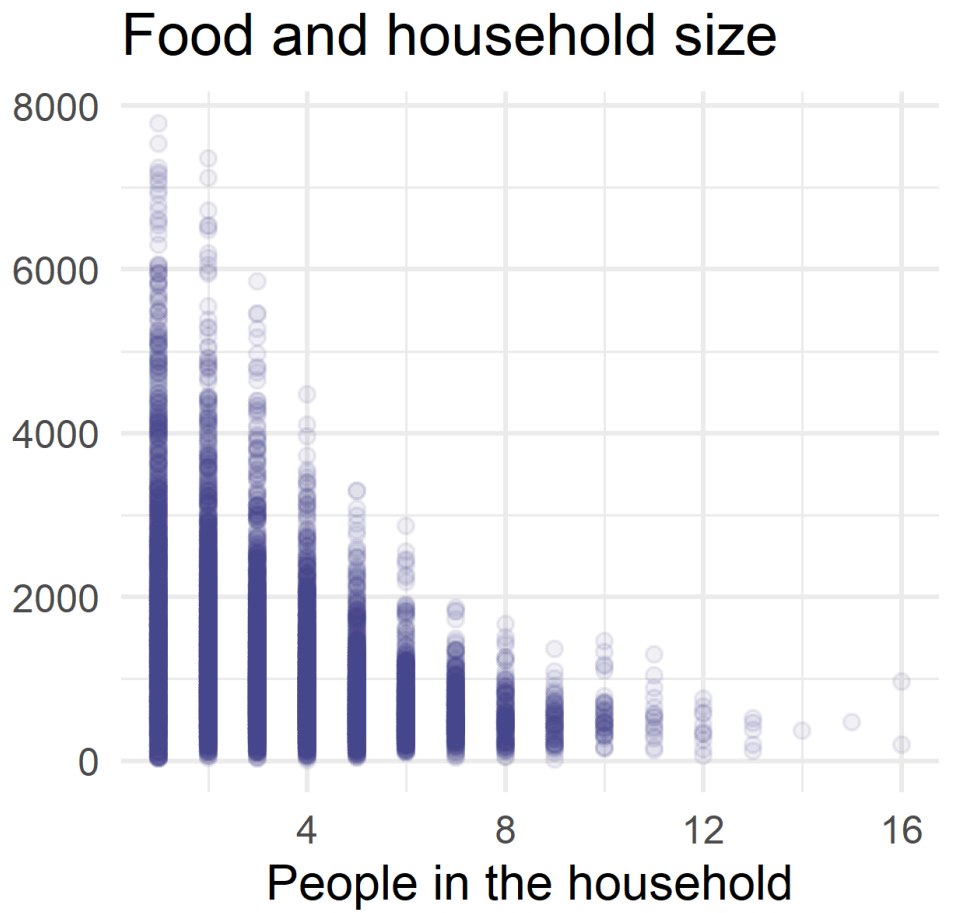
When the cloud is too crowded





Which relationship is stronger?





One cloud rises, the other falls. Larger households spend less per person — cooking for six is cheaper per head than cooking for one.

Covariance

$$\operatorname{Cov}(X, Y) = \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu_X)\,(y_i - \mu_Y)$$

The sample version divides by $n - 1$, exactly as the sample variance does:

$$\hat{\operatorname{Cov}}(X, Y) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

Note that $\hat{\operatorname{Cov}}(X, X)$ is just the variance of X — covariance of a variable with itself.

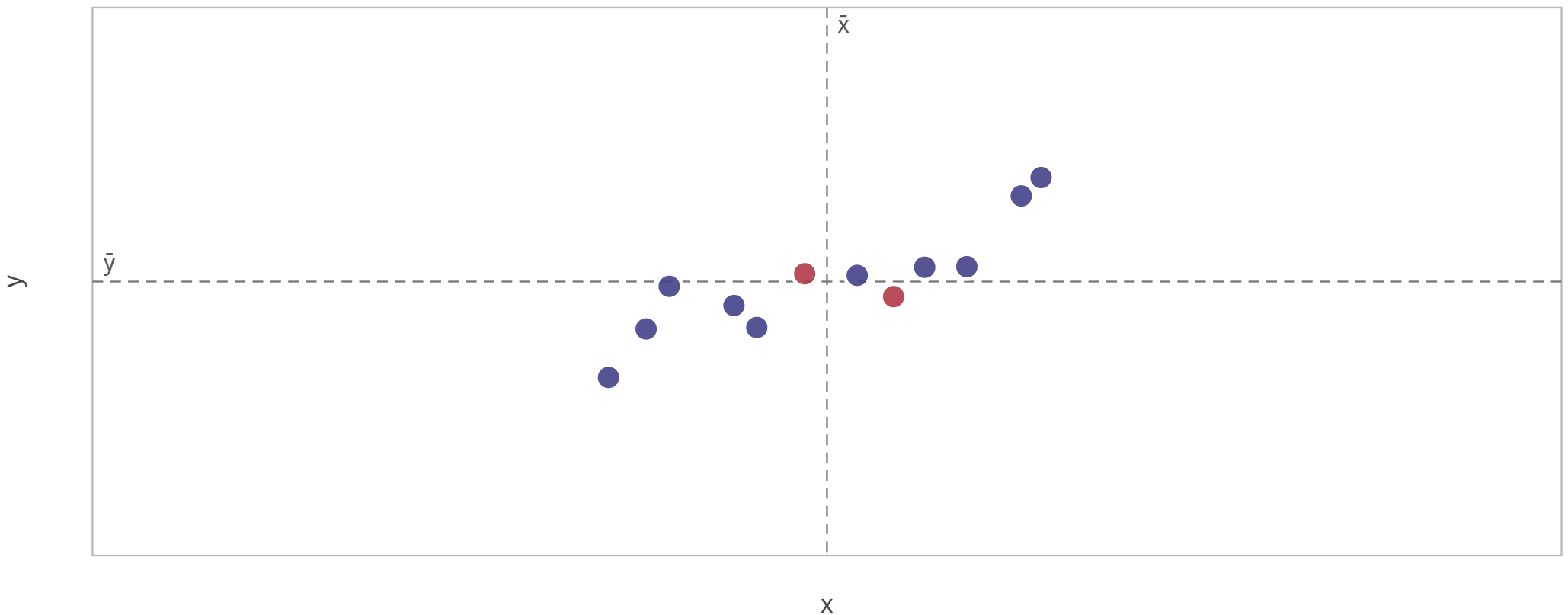
Covariance as rectangles

Each point makes a rectangle: how far it sits from $\bar{x}$, times how far it sits from $\bar{y}$. Purple rectangles count as positive, red as negative. The covariance is their average.

Points 12 Relationship Spread of x Spread of y

Show every rectangle | New points | Show Cov and r

Cov = ? **r = ?** **s_x = ?** **s_y = ?**



Hover a point to see its rectangle. Drag any point to move it. Stretching x or y changes **Cov** but leaves **r** where it was.

What makes a covariance big

- Sign — positive if the two variables tend to sit on the same side of their means, negative if on opposite sides.
- Strength — the more reliably they move together, the more the products reinforce rather than cancel.
- Size of the deviations — a variable measured in bigger units has bigger deviations, and therefore bigger products.

That third one is going to cause trouble.

Exercise 14 — Measure both relationships

Compute the covariance of `spending$food_pc` with `spending$income_pc`, and the covariance of `spending$food_pc` with `spending$members`.

Which pair has the larger covariance?

Exercise | Start Over | Show Hint | Run Code

```
1 # one covariance per line
```

The covariance ranking is not credible

Income wins by a factor of 3,500: $\hat{\text{Cov}} = 2{,}389{,}884$ against -674 for household size.

But look at the two scatterplots again. Neither is 13,000 times tighter than the other. Something is wrong with the comparison.

Incomes per person run into the tens of thousands of pesos ($s = 4{,}597$); household sizes run from 1 to 16 people ($s = 1.8$). Income deviations are simply much bigger numbers, so their products are bigger too — whatever the relationship does.

The two covariances are not even in the same units: one is *pesos squared*, the other *pesos* × *people*. Comparing them was never meaningful.

Covariance depends on the units of both variables, so covariances of different variable pairs cannot be ranked against each other.

Change the units, change the covariance

The survey asked what a household spent on food last month. A yearly figure would do just as well: multiply every household's food spending by 12.

Call the yearly figure $Z = 12Y$. What happens to the covariance with income?

$$\hat{\mathrm{Cov}}(X, Z) = \frac{1}{n-1}\sum(x_i - \bar{x})(z_i - \bar{z})$$

$$= \frac{1}{n-1}\sum(x_i - \bar{x})(12y_i - 12\bar{y})$$

$$= 12 \cdot \frac{1}{n-1}\sum(x_i - \bar{x})(y_i - \bar{y}) = 12\,\hat{\mathrm{Cov}}(X, Y)$$

No household eats differently because you wrote the year instead of the month. Yet the covariance is 12 times bigger.

Exercise 15 — Watch it move

Create `yearly`, the food spending of each household per year instead of per month. Compute the covariance of `spending$income_pc` with `yearly` and compare it to the covariance with `spending$food_pc`.

Exercise | Start Over | Show Hint | Run Code

```
1 # convert, then recompute
2 yearly <- spending$food_pc * 12
```

The general rule

Rescale either variable however you like — multiply by a and c, add b and d. Hover each constant to see what becomes of it.

$$\operatorname{Cov}(aX + b,\ cY + d) = ac \operatorname{Cov}(X, Y)$$

Where it happens — one deviation at a time:

$$(ax_i + b) - \overline{(aX + b)} = ax_i + b - (a\bar{x} + b) = a(x_i - \bar{x})$$

Same idea for the variance: $\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X)$, so $s_{aX+b} = |a|\, s_X$. Adding b does nothing; multiplying by a scales the spread.

Correlation

Divide the covariance by both standard deviations and the units cancel:

$$\rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y}$$

The sample version replaces each piece with its sample counterpart:

$$\hat{\rho}(X, Y) = \frac{\hat{\mathrm{Cov}}(X, Y)}{s_X \cdot s_Y}$$

Because the units cancel, ρ always lands between -1 and 1 — and correlations of completely different variable pairs can be compared.

Exercise 16 — Rank them properly

a. Compute the correlation of `spending$food_pc` with `spending$income_pc`, and with `spending$members`. How many times stronger is the winner — and what is the sign of each?

b. Recompute the income correlation using `yearly` from Exercise 15. What happens to it?

Exercise | Start Over | Show Hint | Run Code

```
1 yearly <- spending$food_pc * 12
2
3 # (a) two correlations
4
5 # (b) the same one, with yearly food spending
```

Why correlation does not move

$$\hat{\rho}(X, Z) = \frac{\hat{\mathrm{Cov}}(X, Z)}{s_X \cdot s_Z}$$

Scaling Y by 12 multiplies the covariance by 12 — but it also multiplies s_Z by 12:

$$= \frac{12 \cdot \hat{\mathrm{Cov}}(X, Y)}{s_X \cdot 12\, s_Y}$$

$$= \frac{\hat{\mathrm{Cov}}(X, Y)}{s_X \cdot s_Y} = \hat{\rho}(X, Y)$$

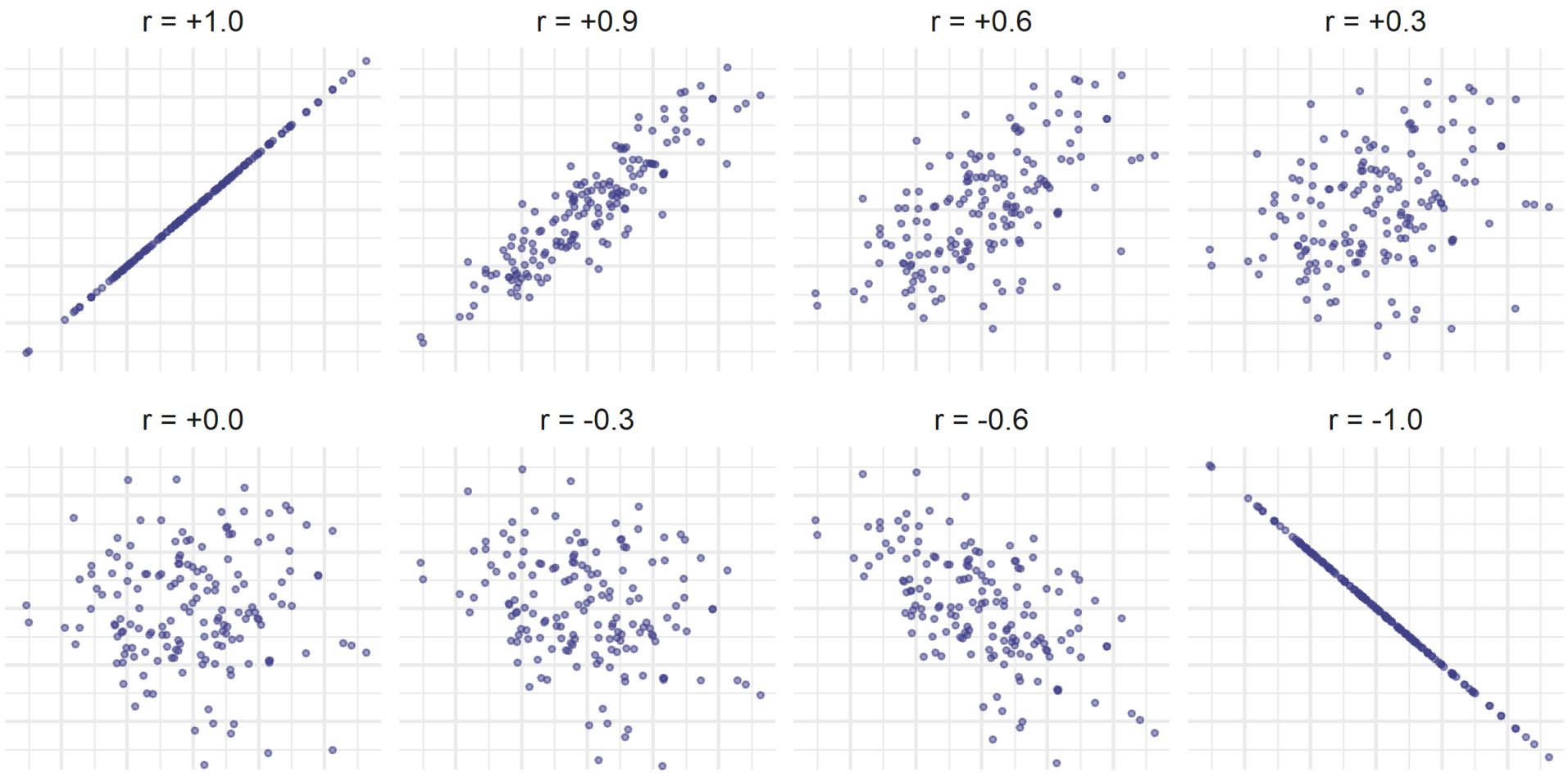
The 12 cancels top and bottom. Correlation is scale-free — which is exactly why it, and not covariance, is the number you report.

In general, for any constants a, b, c, d:

$$\rho(aX + b,\ cY + d) = \rho(X, Y) \quad \text{if } ac > 0$$

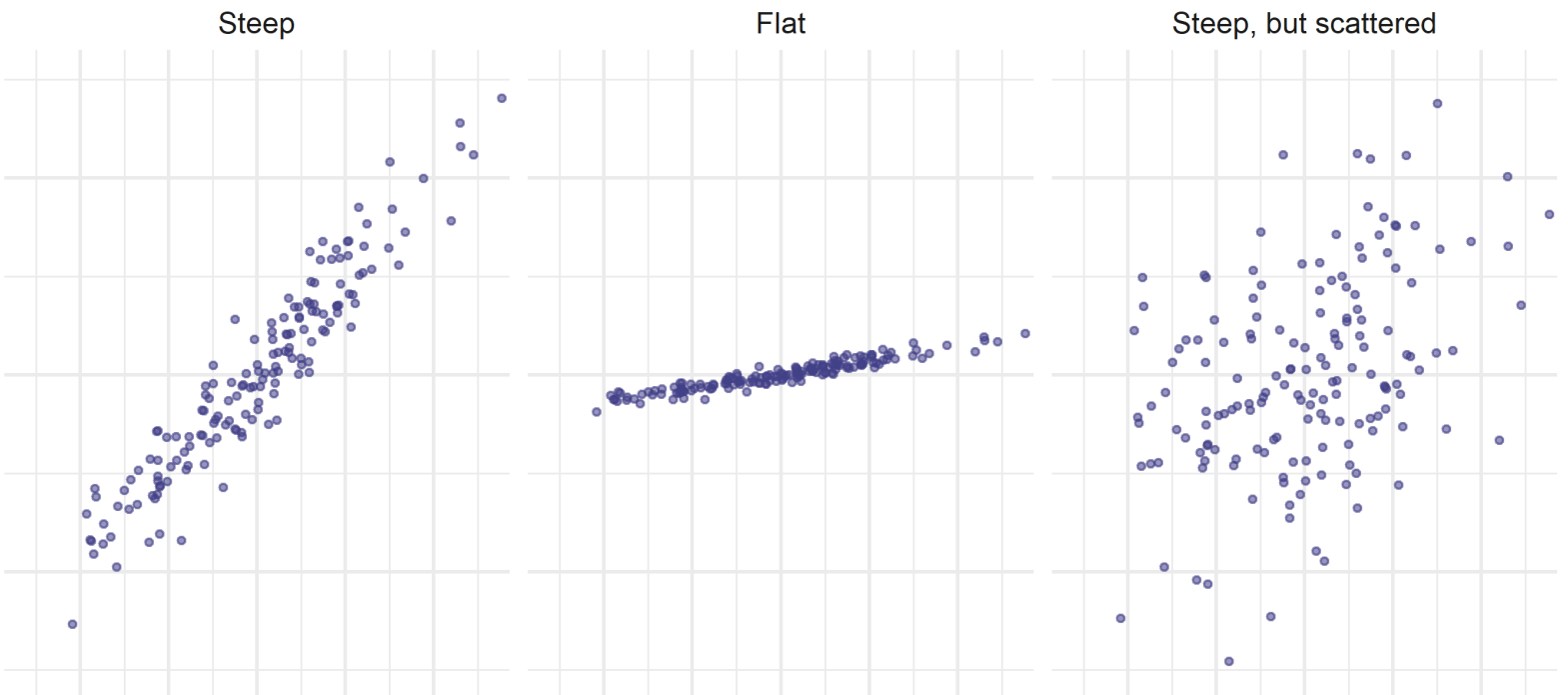
Correlation ignores all four. Its size never changes; only its sign flips, and only if exactly one of a and c is negative.

What different correlations look like



- Every panel has the same points on the horizontal axis and the same axes. Only the correlation changes.
- Read left to right: a perfect line, then the cloud fattens as the correlation falls to zero, then it tightens again in the other direction.
- The sign says which way it leans. The size says how tightly the points hug the line.

Flat does not mean weak

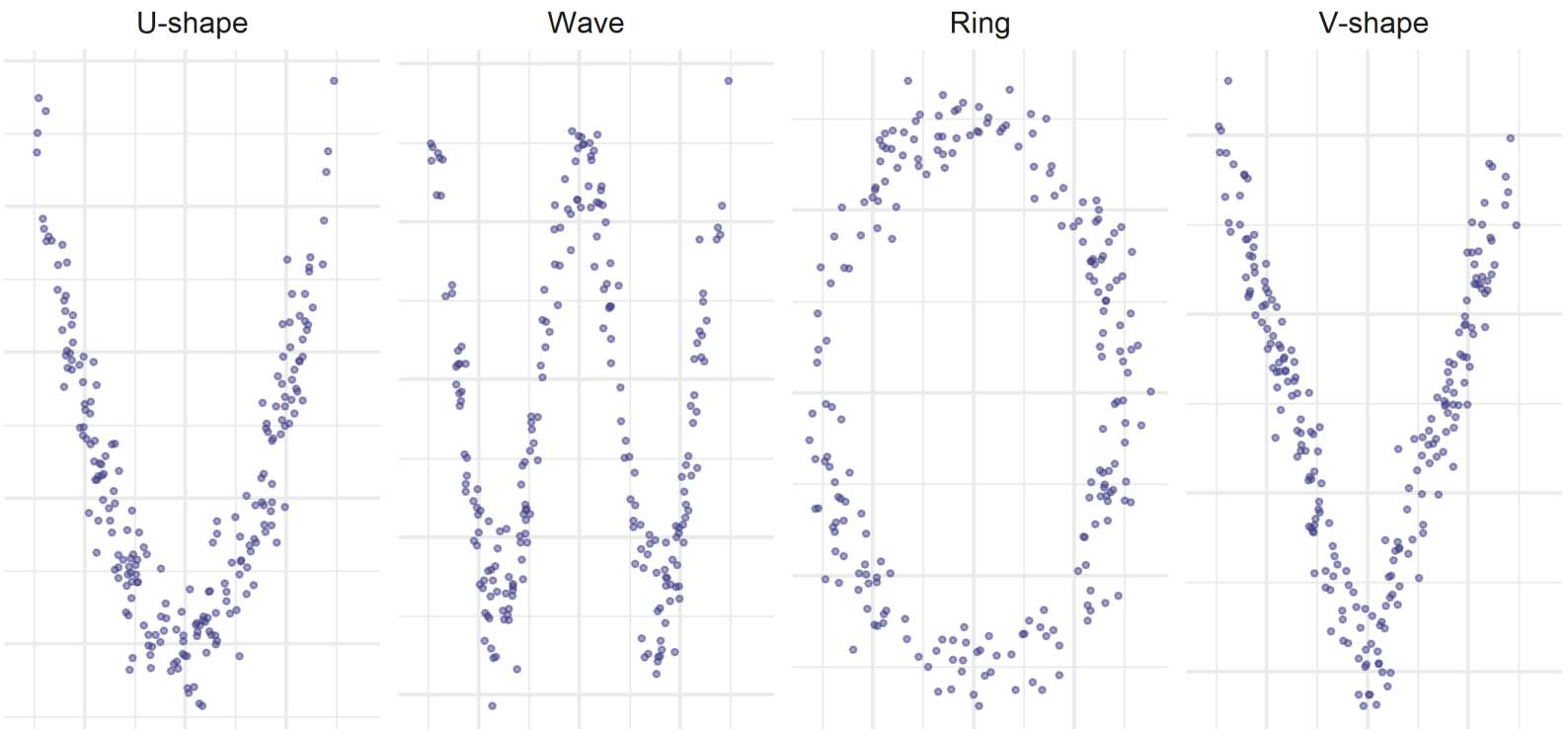


Steep — ? **Flat — ?** **Steep, but scattered — ?**

- Which two have the same correlation? Hover to check.
- The first two panels have the same correlation. The second is far flatter — its y values were simply multiplied by 0.15 — and r did not move.
- The third panel is as steep as the first but the points are scattered, and r drops to 0.40.
- So steepness is not strength. Correlation answers “how close to a line?”, never “how steep is the line?” — the slope is a different question, and it is what regression answers in Week 8.

Correlation only sees straight lines

Four data sets, each with an obvious pattern. What is the correlation in each one? Decide first, then hover a name below the picture to check.



U-shape — ? Wave — ? Ring — ? V-shape — ?

- Knowing x tells you a great deal about y in every panel. A correlation near zero says only that no straight line fits.
- So $\rho = 0$ does not mean the variables are unrelated, and it does not mean they are independent. It means *no linear association*.
- Correlation is one particular kind of association. Always draw the picture first.

Five things to remember

1. Correlation always lies in $[-1, 1]$: $-1 \leq \rho(X, Y) \leq 1$.
2. $\rho = 1$ means the points sit exactly on an upward line; $\rho = -1$, exactly on a downward line.
3. $\rho = 0$ means no linear relationship. It does not mean the variables are independent.
4. Correlation only sees straight lines — see the next slide for a relationship it misses completely. Always look at the picture before trusting the number.
5. Correlation says nothing about causation.

A relationship correlation cannot see

$$\rho(\text{age of head},\ \text{food per person}) = -0.009$$

By that number there is no relationship at all. Now bin the data.



- Spending per person is high for young heads, falls to about 1,030 pesos in the early forties, climbs back to 1,236 by the late fifties, then falls again after 70.
- Every up-slope cancels a down-slope in the correlation formula, so ρ lands on zero. The relationship is real; correlation simply cannot represent it.

Correlation is not causation

LED bulbs	Mean monthly bill
No	142.9
Yes	196.8

LED bulbs use less electricity than incandescent bulbs. Yet in `households`, the homes with LED bulbs pay the higher bills — by about 54 pesos a month.

Does that mean installing LED bulbs raises your bill?

Exercise 17 — Find the confound

In `households`, compare LED and non-LED homes on something other than the bill — `household_size`, for example — and use it to explain the puzzle.

Exercise | Start Over | Show Hint | Run Code

```
1 # what else differs between the two kinds of household?
```

The less obvious cases

A media campaign. You look at historical data and find that people who saw more ads were less likely to buy. Should you cut the ad budget?

Ask who got targeted. If the campaign deliberately went after weak prospects, the ads were sent *because* purchase probability was low. The correlation is running backwards.

Education and income. They correlate strongly — that is a fact about the data. Does it follow that *you* will earn more by getting a degree?

Only if the people who get degrees are otherwise like the people who do not. They usually are not.

What settles it

- Spurious correlations are everywhere — with enough series, something always tracks something else (tylervigen.com/spurious-correlations).
- No amount of extra data turns a correlation into a cause.
- What does: an experiment that assigns the treatment at random, so the two groups cannot differ systematically beforehand.

That is the A/B test, and it is where Week 7 ends up.

← Back to the recycling table

Exam Spotlight

The real questions — and how we solve them.

→ More exam practice at the end of this deck

Exam question — Fall 2025 Midterm 1

Real question from last year's Midterm 1 — Households & Environment Survey (each household surveyed once, 2017). We solve this together in class.

Q3 (Distributions). *An energy company wants to understand the distribution of weekly household gasoline spending to design a fuel discount program.*

a. Visualize the distribution of `households$gasoline_spending_weekly`.

b. Describe its shape — modality and skewness.

The real dataset is loaded as `households` (2,000 households). Solve it here, exactly as you would in the exam.

Exercise | Start Over | Show Hint | Run Code

```
# a) the right plot for a numeric variable:

# b) numbers that back up your description of the shape:
```

Exam question — Fall 2025 Midterm 1, Q4

Q4 (Associations). *A transport consultant needs to know what drives gasoline spending.* Which is more strongly associated with weekly gasoline spending — the number of vehicles or the household size? Choose an appropriate metric, compute it, and interpret.

Would the conclusion change if spending were in USD instead of MXN (1 USD = 17 MXN)?

Same `households` data. Solve it here, exactly as you would in the exam.

Exercise | Start Over | Show Hint | Run Code

```
1 # 1. pick the metric and compute it for both candidates
2
3 # 2. redo it with spending converted to USD
```

Exercise 6 — The whole week, one question

You work for the weight-loss subscription start-up. The company wants to sell first to people with diabetes, and has to decide whether that group needs its own product.

Compare `Health_data$weight` for the two groups in `Health_data$diabetes` (0 = no diabetes, 1 = diabetes):

a. Is the median weight higher for people with diabetes? By how many kilograms?

b. Is the spread of weight similar in the two groups?

c. Is the difference big enough to justify a separate product? One sentence.

Exercise | Start Over | Show Hint | Run Code

```
# (a) the median of each group

# (b) the spread of each group -- pick a measure and say why

# (c) one plot that shows both groups at once
```

What you must be able to do this week

- Explain why a frequency table works for a variable whose values repeat and collapses for a continuous one.
- Build a histogram, choose the number of intervals, and defend the choice — and build the frequency table behind it (n_i, p_i, P_i).
- Read a histogram and describe the distribution's shape — modality (uni-/bi-/multimodal) and skewness (which tail is longer).
- Use the mean and the median together to say which way a distribution is skewed, without looking at the picture.

Next week: from describing data to reasoning about uncertainty — estimators and the Central Limit Theorem.

Practice for this week

Weekly exercise set: `W03_exercises.pdf`, with solutions in `W03_solutions.pdf`.

You can already solve exercises 1–7, 9 and 10.

Not in these slides: exercise 8, the variance of a portfolio of two stocks (it needs the variance of a sum).

Suggested before the next class, closest to the exam:

Exercise	Topic	How
6	Price, bidders and condition in eBay auctions (`auctions`)	R with the data
5	Marital status, wages and race (`cps`)	R with the data
9	Hours worked, hourly wage and education (`cps`)	R with the data
10	The shape of the income distribution, ENIGH	R with the data
7	Units of the mean, variance, covariance and correlation	R as a calculator

Appendix — optional (extra & theory)

Extra — Does it run in the family?

A contingency table on a second dataset. In `Health_data`, `Mother_diabetes` is `1` if the person's mother had diabetes and `diabetes` is `1` if the person does. Cross-classify the two, rescale within each row, and report the two rates.

Exercise | Start Over | Show Hint | Run Code

```
1 # the same two steps, on Health_data
```

Behind every histogram there is a table

The exam asks for the table, not the picture: interval, midpoint, absolute frequency n_i, relative frequency p_i, and cumulative frequency P_i.

Build it before you see ours.

Exercise 7 — Build the frequency table

Cut `Health_data$weight` into 5 equal intervals — you already have the edges from Exercise 2b — and produce the five columns, one interval at a time. Then answer: what share of people weigh 94.55 kg or less?

The five intervals run 30.37 – 62.46 – 94.55 – 126.63 – 158.72 – 190.81. Build each edge from the minimum and the width rather than typing it, so nothing is lost to rounding.

Exercise | Start Over | Show Hint | Run Code

```
1  # 1. the five edges, built from min and width
2
3  # 2. n_i: how many people fall inside one interval
4  # 3. p_i: what share of the sample that is
5  # 4. P_i: the share at or below the interval's upper edge
```

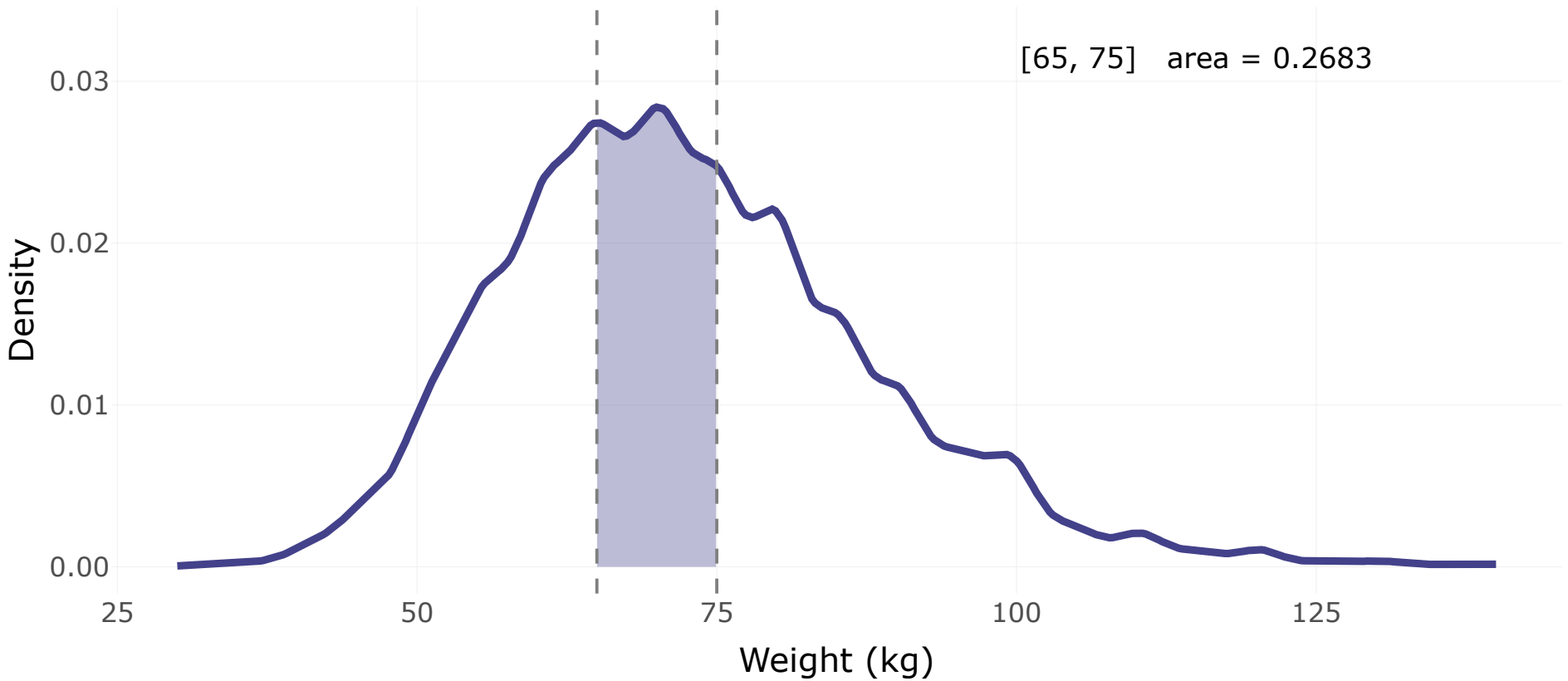
Histogram with 5 classes: the table

Interval	Midpoint	n_i	p_i	P_i
30.37 - 62.46	46.42	10068	0.266	0.266
62.46 - 94.55	78.50	24430	0.645	0.911
94.55 - 126.63	110.59	3206	0.085	0.996
126.63 - 158.72	142.68	143	0.004	1.000
158.72 - 190.81	174.76	11	0.000	1.000

PDF: shading the area, range by range

Same curve as in the main flow — the density of `Health_data$weight`. Each picture prints the area of the shaded stretch, which is the probability of landing in it.

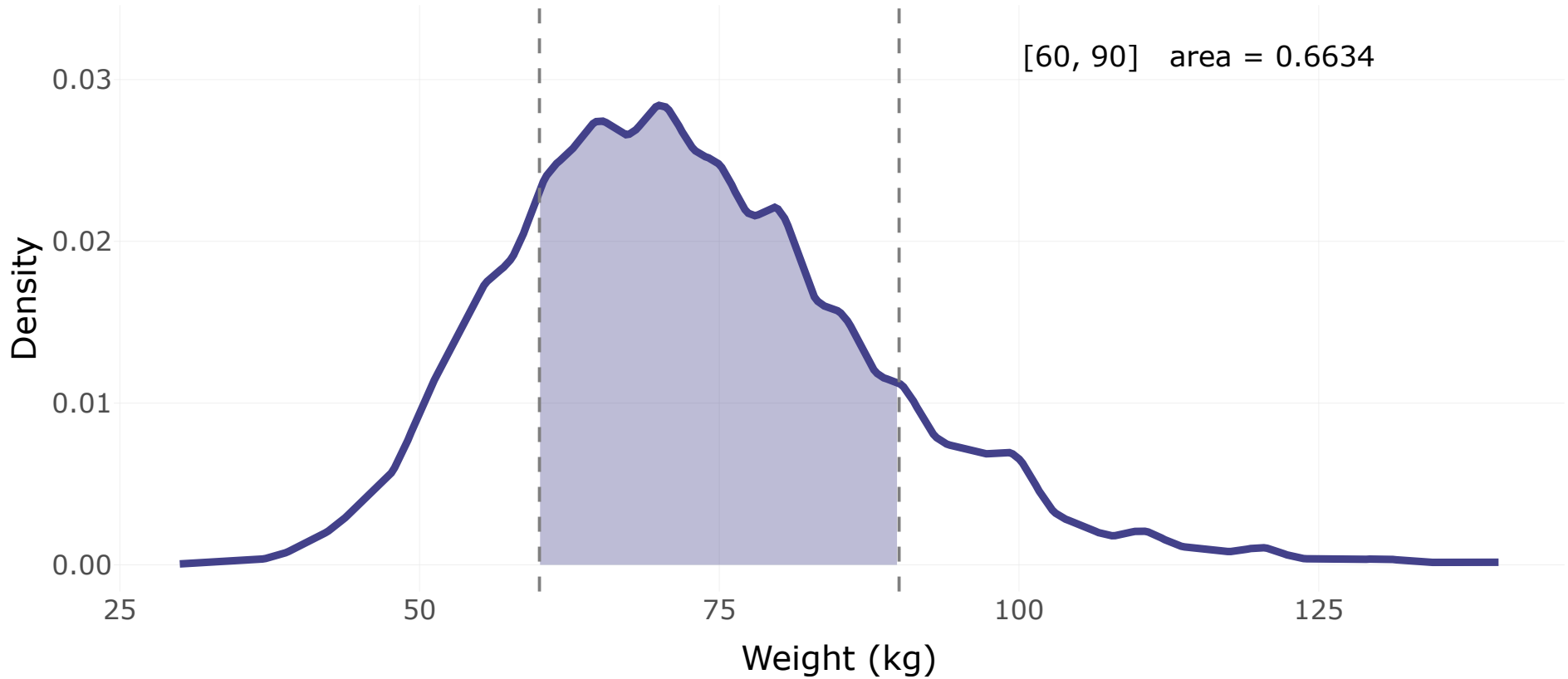
What share of the population weighs between 65 and 75 kg?



27% — the same shaded area you saw on The PDF — where the values pile up.

PDF: a wider range

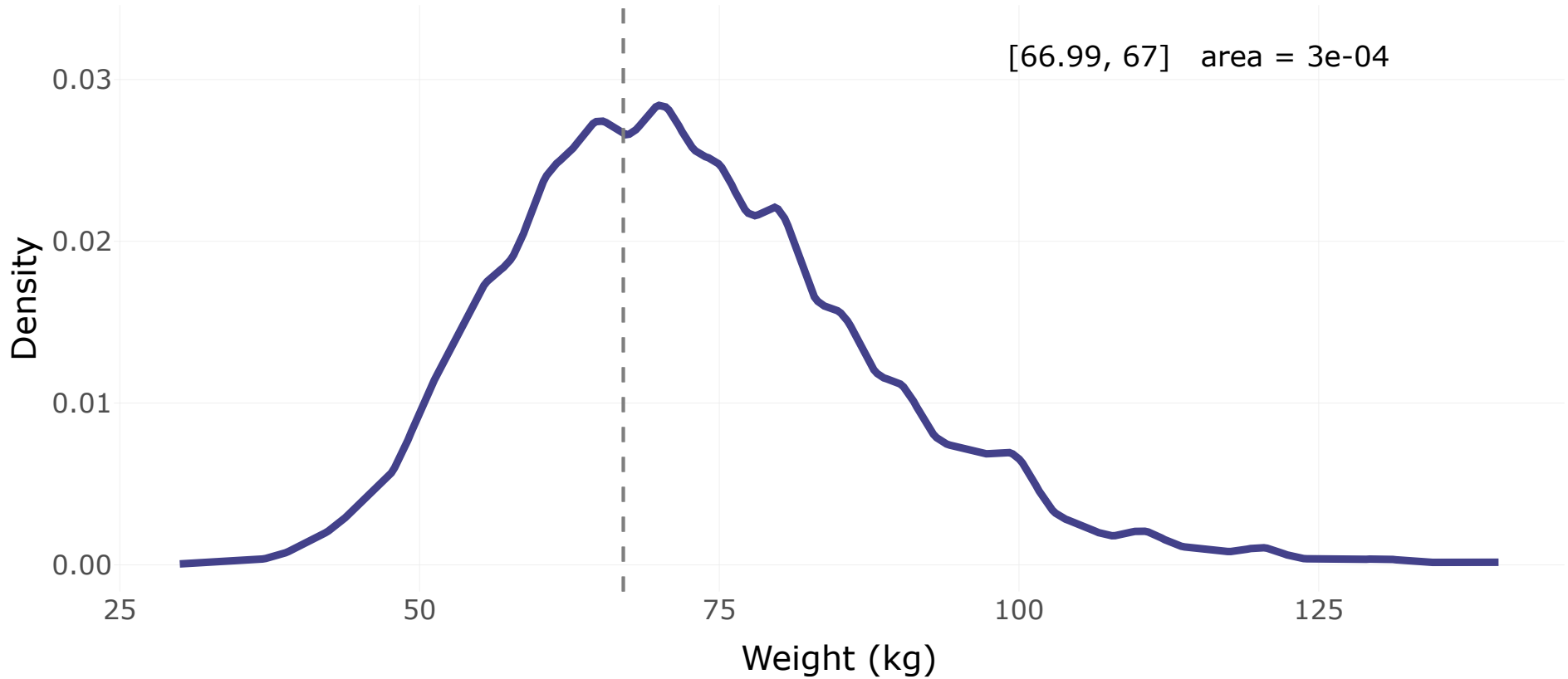
What share of the population weighs between 60 and 90 kg?



66% — a wider stretch means more area, so a larger probability.

PDF: a razor-thin range

What share of the population weighs between 66.99 and 67 kg?

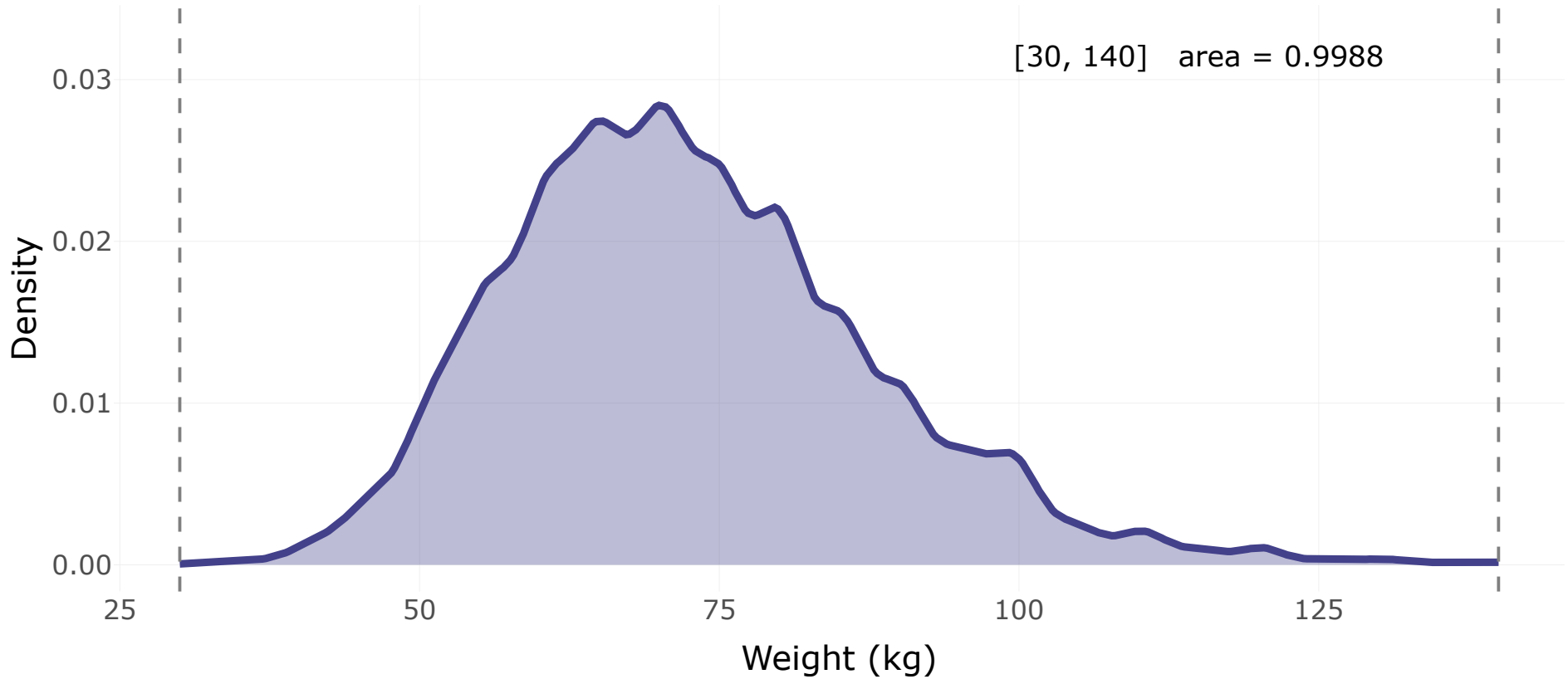


0.0003 — squeeze the stretch to a single point and the area goes to zero. That is why the pdf gives densities, not probabilities.

← Back to The PDF — where the values pile up

PDF: the whole range integrates to 1

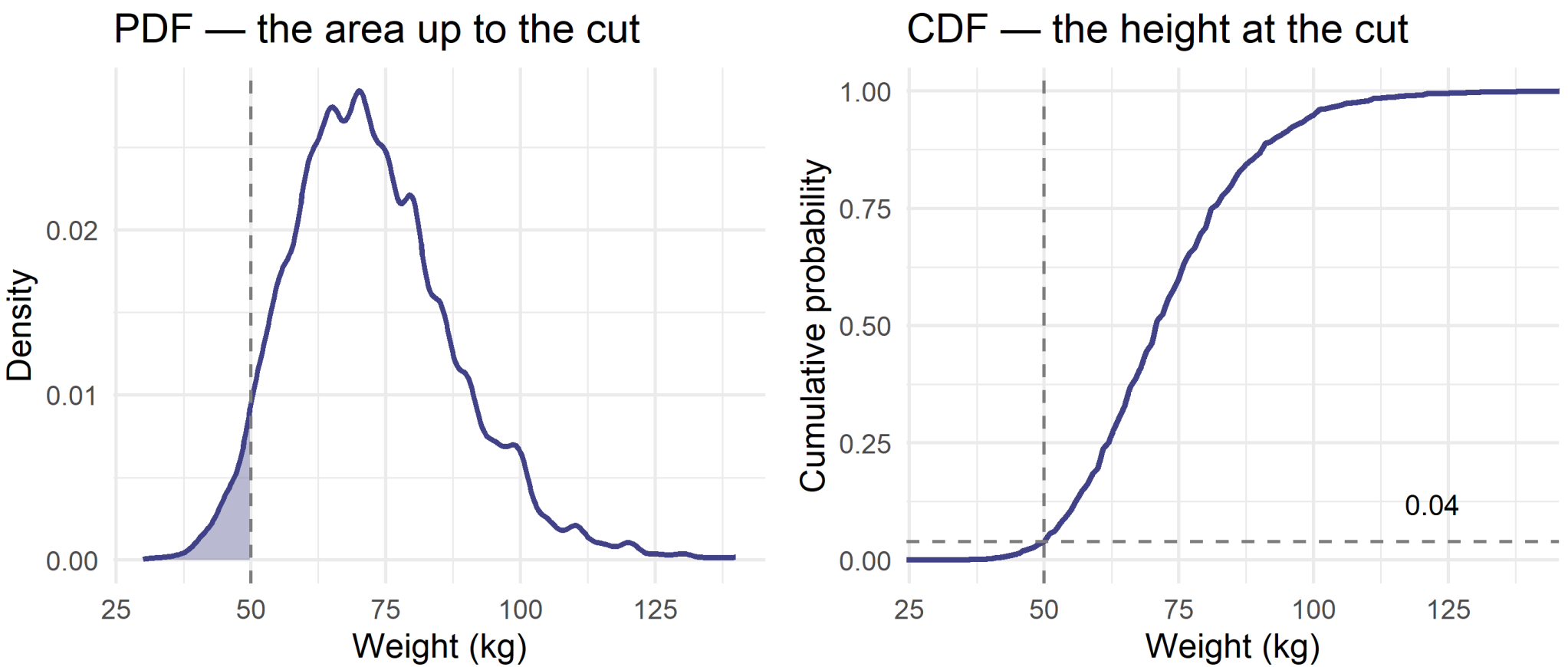
What share of the population weighs between 30 and 140 kg? (essentially everyone)



- The area under the entire curve is 1.

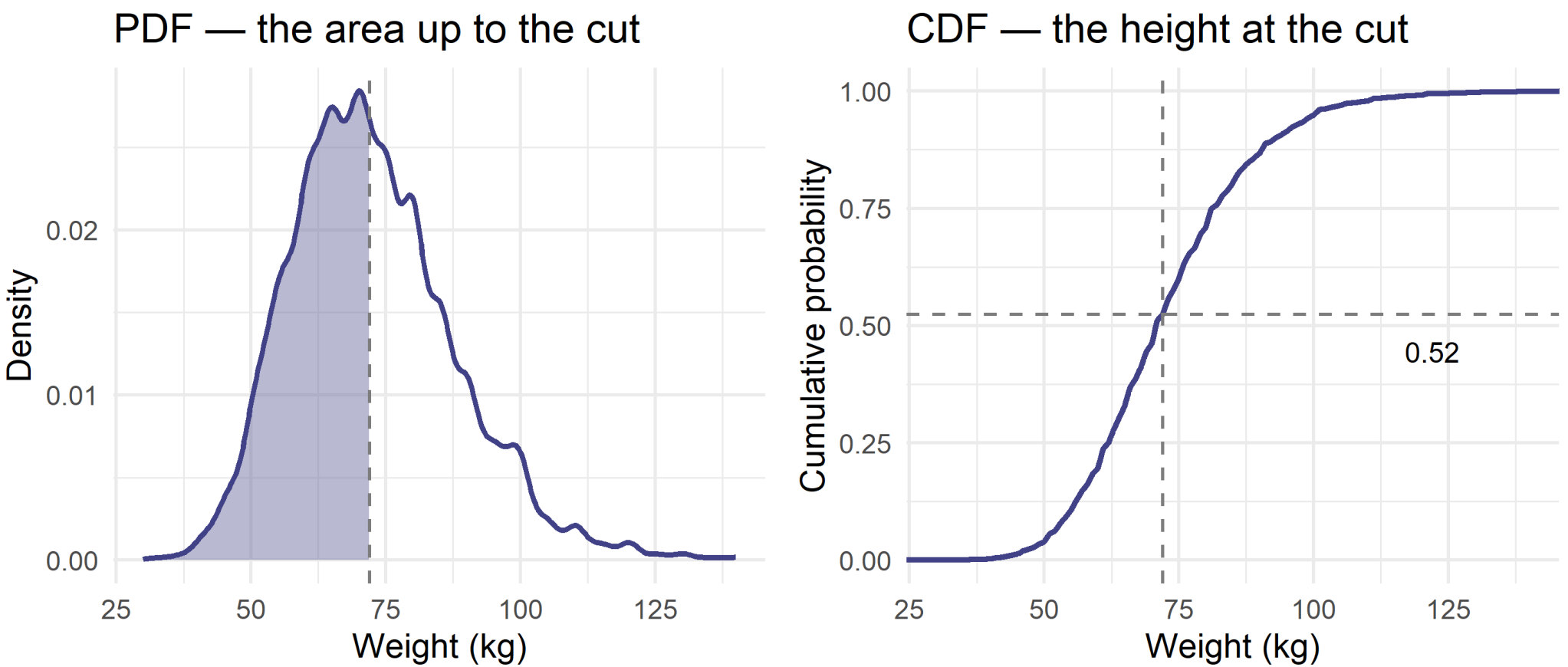
CDF Example 1: weight in the population

$$F(50) = P(X \leq 50) = 0.04$$



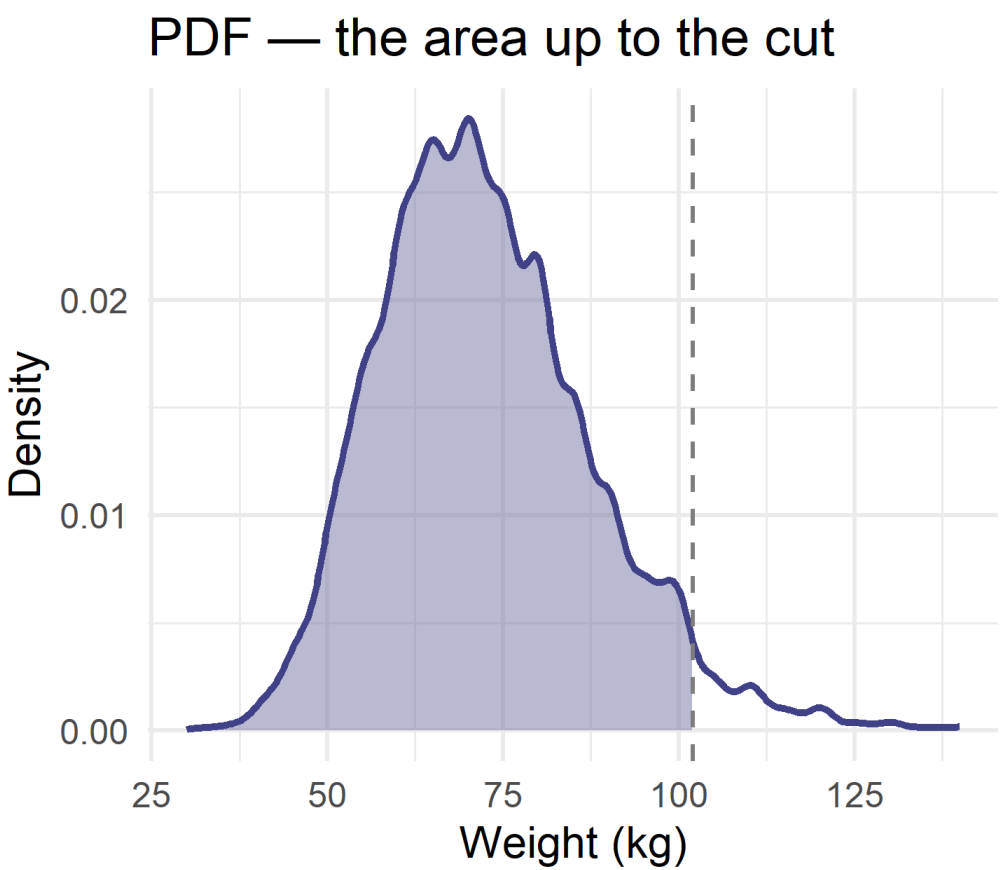
CDF Example 2: weight in the population

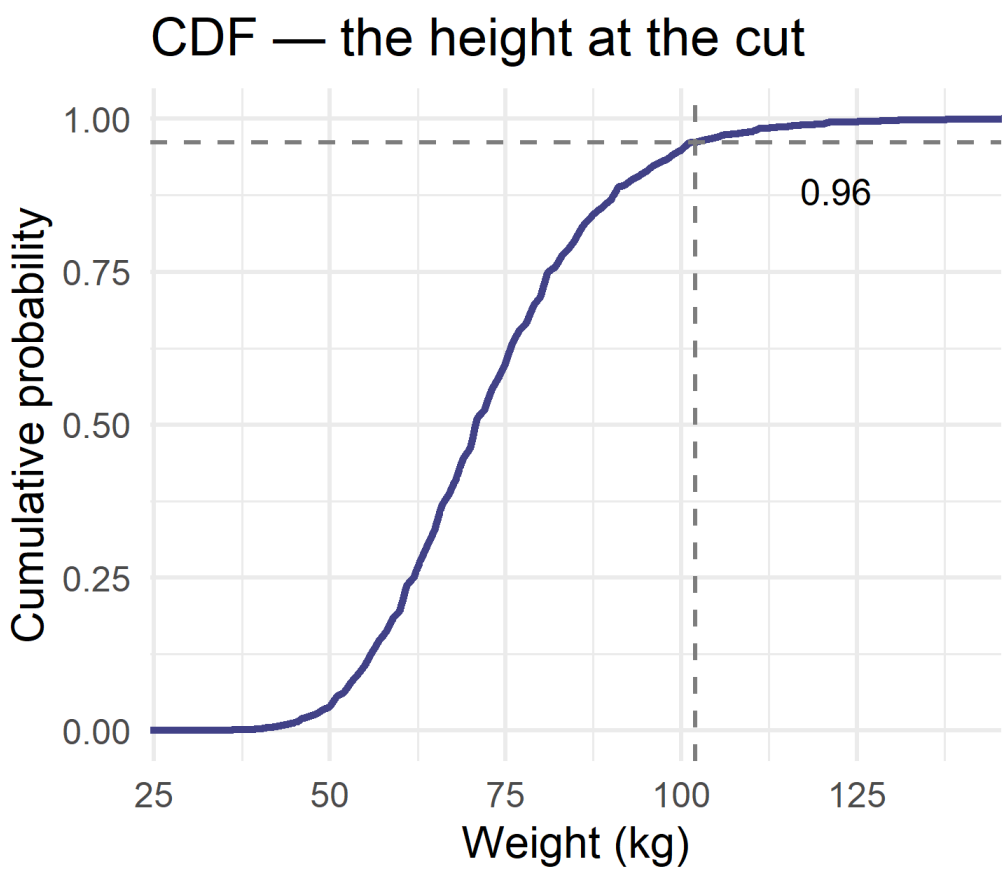
$$F(72) = P(X \leq 72) = 0.52$$



CDF Example 3: weight in the population

$$F(102) = P(X \leq 102) = 0.96$$

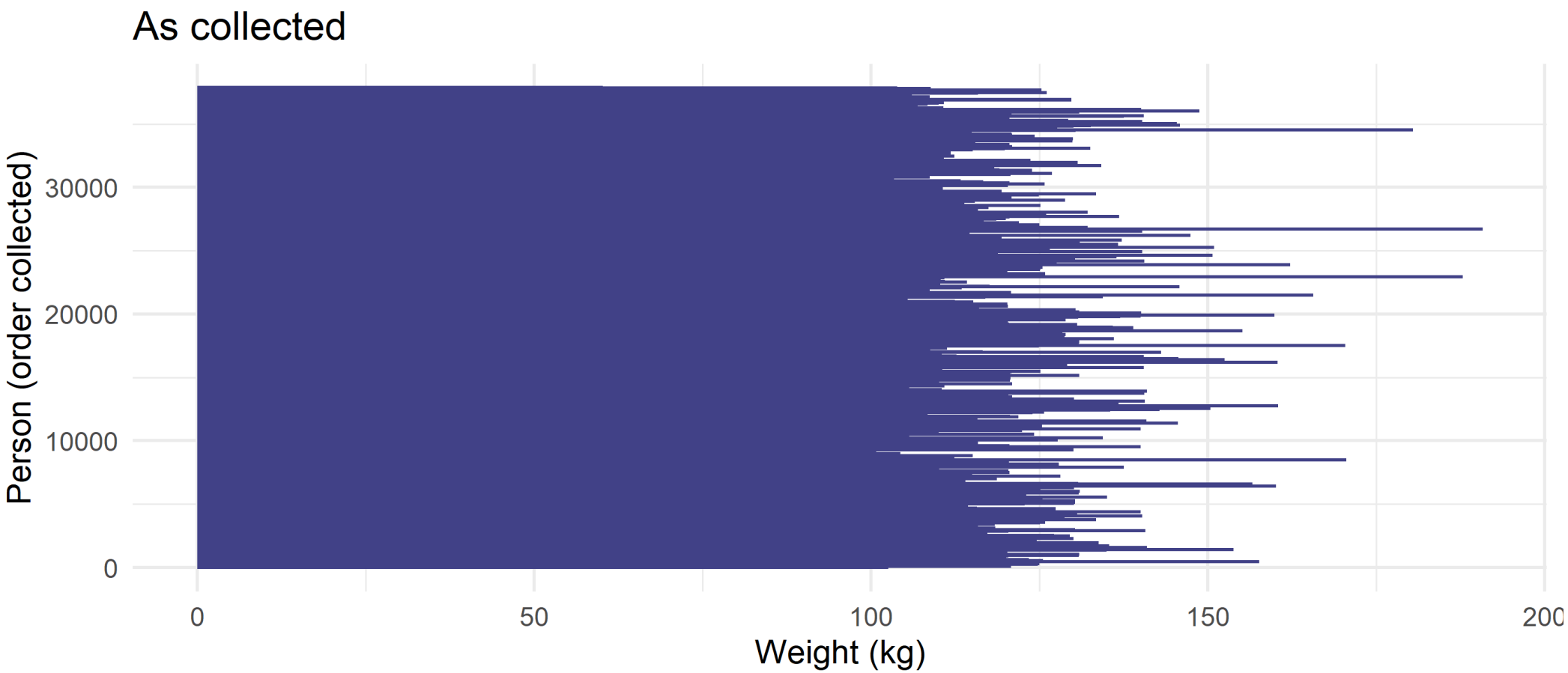




Never integrate a CDF! (it is already the integral of the pdf)

Empirical CDF

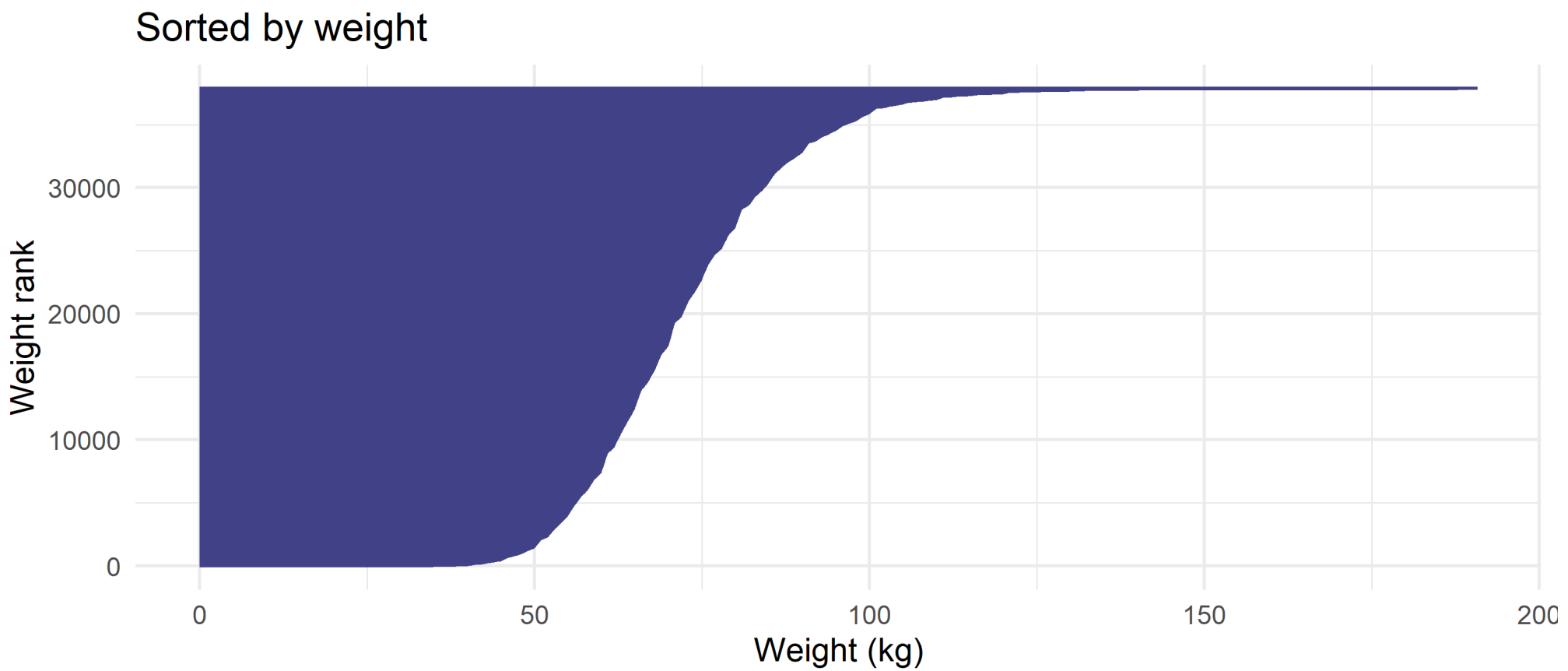
What if we only have a sample and don't know the true pdf? Start with the sample exactly as it arrives — one bar per person, in the order they were surveyed.



- Every weight in the sample is on this picture, and it says nothing.
- Survey order is arbitrary, so the ragged edge carries no information.

Empirical CDF: now sort it

Same people, same weights — lined up from lightest to heaviest.



- Nothing was added — only the order changed — and a curve appears.
- Each person's position on the vertical axis is now their rank: how many people weigh the same or less.

Turn that picture on its side and divide the rank by N — that *is* the curve from The CDF — how much is at or below.

Empirical CDF (cont.)

$$ECDF(x) = \frac{\sum I(w_i \leq x)}{N} = \frac{\text{Number of people with weight} \leq x}{N}$$

$I(w_i \leq x) = 1$ if person i's weight is $\leq x$ (*indicator function*); N = sample size.

Evaluate that at every x and you get a curve. What does it look like?

Empirical CDF: finding percentiles

- Say we want the 25th percentile (1st quartile), call it v_{25}
- By definition $P(X \leq v_{25}) = 25\%$
- So the 1st quartile is the value where the CDF equals 25%
- More generally, percentile Z: $P(X \leq v_Z) = Z\%$

🖊️ Exercise 8 — Read a percentile off your own ECDF

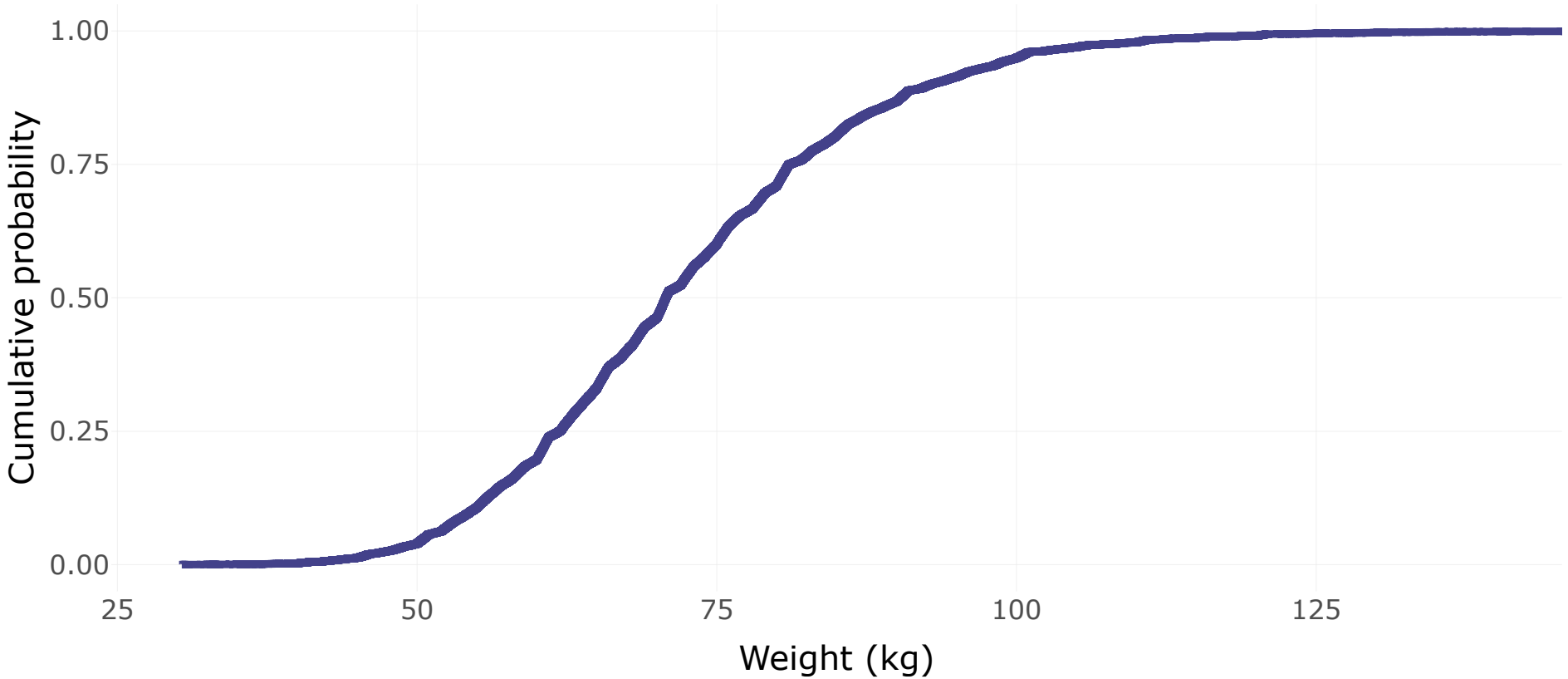
Build the ECDF of `Health_data$weight` yourself. Find the height 0.25 on the vertical axis, trace across to the curve and down to the axis, and read the 25th percentile off your picture — then check the number with `quantile()`.

Then use the curve in the other direction: what share of people weigh 100 kg or less?

Exercise | Start Over | Show Hint | Run Code

```
1 e <- ecdf(Health_data$weight)
2 # 1. plot it, 2. read the weight sitting at height 0.25, 3. check it,
3 # 4. then the share of Health_data$weight at or below 100
```

Empirical CDF: the curve



- Pick a height and read across: height 0.25 lands on q_1.
- Pick a weight and read up: the height above 50 is the share weighing 50 kg or less, $P(\text{weight} \leq 50) = ECDF(50)$; the share above 100 kg is $1 - ECDF(100)$.

← Back to The CDF — how much is at or below

Appendix — scenario & population vs sample

This week: distributions, fast and hands-on

- This is review — you've met distributions before. We move quickly and do it in R.
- What a business analyst needs to *read* a distribution: histograms and shape (modality, skewness).
- We also go a bit deeper on the theory: the PDF and CDF sections.
- Behind on the underlying stats? Study those sections; if you're solid, skim them.
- Practice as we go with the exercise slides embedded in this deck.

Summary

- In the last class:
  - We discussed the organization of the course
  - We overviewed forecasting methods
  - We learned about methods of qualitative forecasting
  - *Reference:* Forecasting Methods and Applications, chapter 1
- This set of classes:
  - We will start learning about exploratory analysis preparing the forecast
    - We will learn to translate business problems to statistical problems
    - We will learn about various data types
    - We will learn how to summarize data and observed patterns
    - *Reference:* Forecasting Methods and Applications, chapter 2.1-2.4

Why do we care?

- To use scientific methods we need to set business problem as a statistical problem
- Define thins that we want to find to identify appropriate methods

Specific Scenario

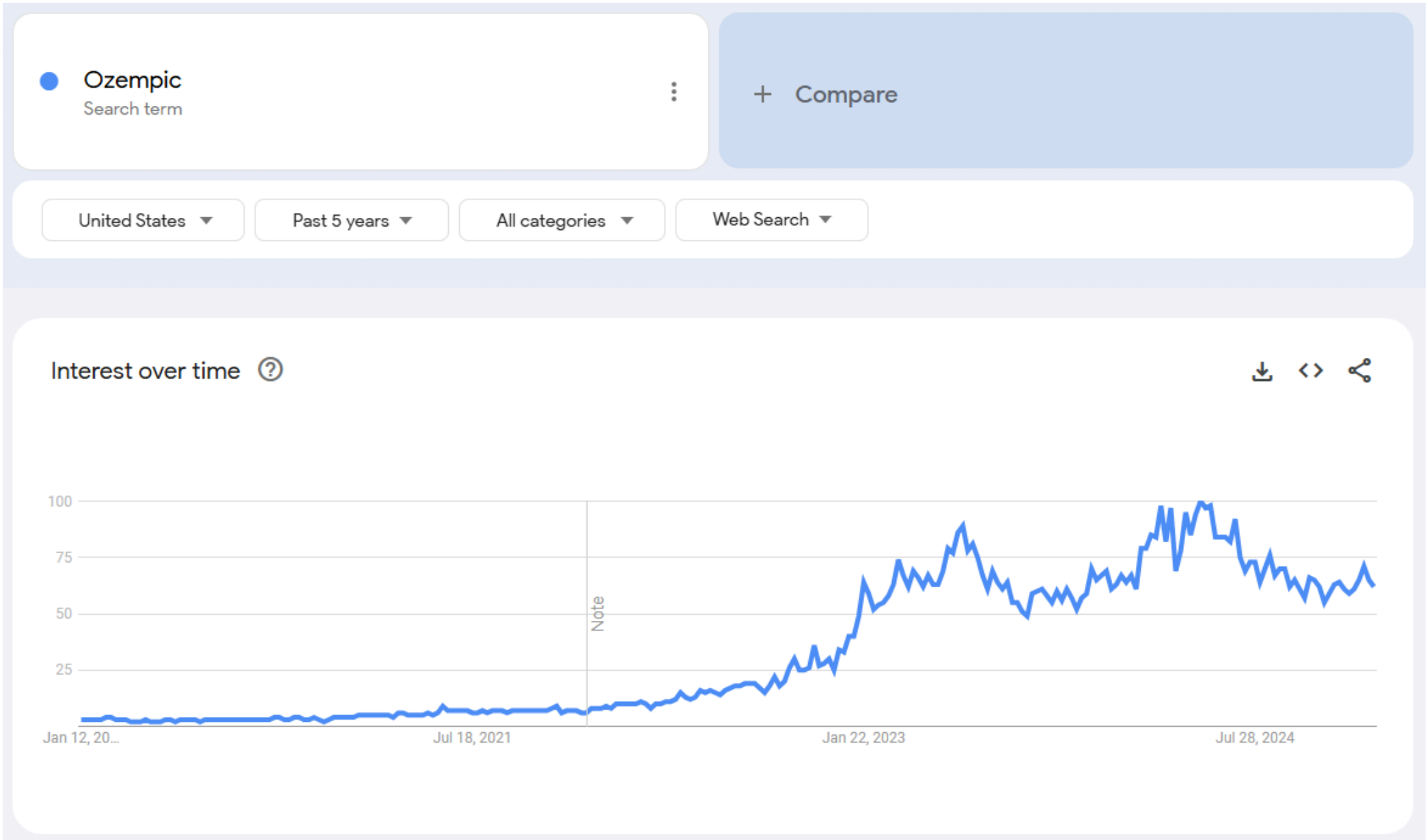
The Opportunity

- Online pharmacies are rapidly growing, offering convenience and accessibility to customers.
  - Example in Mexico: *Choiz*, which provides prescription services and drug subscriptions.

The Trend

- A new generation of highly effective anti-diabetes medications, like *Ozempic*, is gaining popularity for dual benefits: managing diabetes and aiding weight loss.

Search Interest in Ozempic



Your Role

- You have been hired as a consultant for a start-up aiming to launch a subscription service for these medications in Mexico.



The Challenge

- Your boss needs a detailed exploratory market analysis to forecast sales and identify key customer segments

Exam practice

This week's exercise-set problem on real data — run the commands, look at what comes back, then decide. Press Solution once you have your own answer.

← Back to the Exam Spotlight

Exam practice 1 — Website arrivals

Use the `website` dataset (loaded for you), a time-series dataset containing visitor arrivals. [D7]

a. Draw a histogram of `website$arrival`. How would you describe the modality of this distribution?

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```